

SPECTRUM: A Data-Driven, Component-Level Projection Framework for 21st-Century Sea-Level Rise

Minchew, Meyer, et al.

Abstract

Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models (SPECTRUM)

Every centimeter of sea-level rise exposes millions more people to annual coastal flooding, yet projections of sea-level come with large uncertainties. Many of the commonly used projections rely on complex models calibrated on data too sparse to adequately constrain them, causing the outputs to depend on model assumptions in ways that remain unquantified and nontransparent. Here, we construct a data-driven sea-level forecast by calibrating parsimonious physical models for individual sea-level contributors against decades-long independent observational records. When forced with observed surface temperatures, the calibrated components sum to match observed global mean sea level rise (GMSLR) from satellite altimetry collected since 1993. Hindcasts match GMSLR reconstructions within error to at least 1950, or approximately 1 °C warming. Our forecasts show a 69% probability of GMSLR exceeding 1 meter, and 5% of exceeding 2 meters, by the end of the century. The dynamics of West Antarctic glaciers account for $\sim 90\%$ of the forecast uncertainty and reveal orthogonal dimensions of sea-level risk: emissions reduction decreases median projections related to the predictable, temperature-sensitive components, but leaves the high-risk tail of the distribution largely unchanged, exposing hundreds of millions of people to accelerating sea-level risks.

1 Introduction

Notes for a new intro:

- overarching motivation; doesn't need to hit as hard bc of audience
- slr projection frameworks — need a scaffolding
- process ice sheets – strengths: detailed representation of physical processes; ability to compare competing and complementary processes; they exist. Weaknesses: super complicated, needs loads of data to constrain, unlikely data will constrain any time soon, opaque, extrapolation beyond calibration bounds
- semi-empirical – strengths: simple and transparent; data-driven; represent quantities of interest. Weaknesses: not physical, not necessarily causal, non-stationary, extrapolation beyond calibration bounds

- intermediate complexity – relatively few. Strengths: parsimony, data-driven (existing data), physical/causal. weaknesses: Lumped parameters (lacks detail); must assume stationary; extrapolation beyond calibration bounds
- value of scaffolding (hierarchy of model complexity) – take advantage of the strengths of different models; simple models act as checks/motivation for more complex models; cal/val within model space; big ideas come from simple models

One of the major challenges of projecting sea-level rise is calibrating, testing, and validating models with data, and then presenting those results with rigorous uncertainty quantification that allows for source separation. A rigorous approach to calibration, validation, and uncertainty quantification is essential because predictions of future sea-level rise are out of sample relative to the modern observational record.

Many projections of sea-level rise rely on sophisticated process-based ice sheet models that solve systems of equations representing ice flow, deformation, and fracture. These models have improved dramatically

A sea-level projections framework based on models representing diverse approaches and a hierarchy of complexity with robust testing against data can provide a foundation for reliable sea-level projections and guidance for the development of increasingly sophisticated, physics-based models that ***value for sea-level rise projections and to help guide the development of models that allow for deeper understanding of the salient physical processes. ***Finish paragraph on the value of different types of sea level forecast models***

At the highest level of sophistication are the numerical ice sheet models, including ISSM, PISM, MALI, that represent the dynamics of ice flow and fracture, and the interactions of ice sheets and the broader climate system. These models rely on a variety of parameterizations for different physical processes including ice-ocean interactions, iceberg calving, the sliding of ice over bedrock and sediment, the hydrology...*** In their current state, process-based models, diverge from historical records in hindcasting studies (*Aschwanden et al.*, 2021) and from satellite observations within a decade of initialization (*Fricker et al.*, 2025). The reasons for the disagreements between observations and ice sheet models are manifold and described in greater detail in other publications ***. In essence, the disparities arise from known structural biases, including the absence of leading-order physical processes like iceberg calving (*Fricker et al.*, 2025) and an assumed ice viscosity that is less sensitive to changes in stress than observations indicate (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025), and initialization uncertainties. Decades of general forecasting research shows that models more complex than their calibration data can constrain generally produce worse predictions than simpler models (*Green and Armstrong*, 2015).

The simplest are the data extrapolations, or “naive” forecasts. These approaches use observed GMSL rate and acceleration. The satellite altimetry record provides GMSLR measurements from 1993–2025 that show a constant acceleration of 0.08 mm yr^{-2} ($0.02\text{--}0.14 \text{ mm yr}^{-2}$) and rate 4.5 mm yr^{-1} ($3.5\text{--}5.5 \text{ mm yr}^{-1}$) in 2025 (*Hamlington et al.*, 2022, 2024). ***cite Nerem and other extrapolation papers; ***Extrapolating these values to 2100 yields 66 cm (41–91 cm) of 21st-century rise. Assuming the naive forecast has predictive skill over roughly half the calibration window, the extrapolation of the 32-year satellite record provides predictive skill over the next 15–20 years. Beyond that horizon, the naive forecast is less reliable but nevertheless establishes a scenario-independent, data-driven reference.

The next class of data-driven forecasts are the semi-empirical models that fit the observed sensitivity of the rate of GMSLR to changes in GMST (Fig. 7b). Under current warming trends, Rahmstorf (2007) (*Rahmstorf*, 2007) predicts 0.67 m (0.63–0.71 m) of 21st-century rise, matching the naive forecast and the results of this study within error.

Current coastal adaptation planning relies heavily on projections from the IPCC Sixth Assessment Report (AR6), which estimates 0.4–0.7 m of rise by 2100 relative to the 1995–2014 baseline (*Fox-Kemper et al.*, 2021; *Hirschfeld et al.*, 2023). These projections depend on process-based ice sheet models forced with climate models and compared against one another in efforts such as ISMIP6 ***cite ISMIP6 papers***. Process-based ice sheet models represent the detailed physi

Rates of global mean sea-level rise (GMSLR) have doubled twice in the past 50 years (*Frederikse et al.*, 2020; *Hamlington et al.*, 2024) (Fig. 7). The first doubling took about 16 years and 0.34 °C of warming (*Rohde and Hausfather*, 2020). The second took about 21 years and 0.41 °C of warming (Fig. 7). The rate could double again around 2080 if the current observed sea-level acceleration continues, and by 2040 if it takes no more warming than the previous two doublings. The 40-year gap illustrates the uncertainty that coastal planning and adaptation must absorb.

Every centimeter of sea-level rise exposes roughly 1.6 million additional people to annual coastal flooding (*Kulp and Strauss*, 2019), approximately 90% of whom live in low- and middle-income countries (*Jongman et al.*, 2012). Without additional adaptation, annual flood damages are projected to reach US\$14–18T (trillion) per year for 0.9–1.1 m of rise (*Jevrejeva et al.*, 2018). Estimates of total adaptation costs per unit sea level rise are not available but a lower-bound, rough-order-of-magnitude estimate of US\$4T per meter of sea-level rise is reasonable based on a projected cost of US\$400B simply to raise dikes by 1 meter along only 3.4% of coastlines (*Tiggeloven et al.*, 2020). Developing countries currently receive US\$21B per year in total adaptation finance across all sectors, not just coastal (*United Nations Environment Programme*, 2023), raising questions about the feasibility of coastal adaptation at relevant scales.

Current coastal adaptation planning relies heavily on projections from the IPCC Sixth Assessment Report (AR6), which estimates 0.4–0.7 m of rise by 2100 relative to the 1995–2014 baseline (*Fox-Kemper et al.*, 2021; *Hirschfeld et al.*, 2023). These projections depend on process-based ice sheet models forced with climate models and compared against one another in efforts such as ISMIP6 ***cite ISMIP6 papers***. Process-based ice sheet models represent the physi

The simplest are the data extrapolations, or “naive” forecasts. These approaches use observed GMSL rate and acceleration. The satellite altimetry record provides GMSLR measurements from 1993–2025 that show a constant acceleration of 0.08 mm yr^{-2} ($0.02\text{--}0.14 \text{ mm yr}^{-2}$) and rate 4.5 mm yr^{-1} ($3.5\text{--}5.5 \text{ mm yr}^{-1}$) in 2025 (*Hamlington et al.*, 2022, 2024), already exceeding IPCC AR6 projections in 2025 (Fig. 9a). Extrapolating these values to 2100 yields 66 cm (41–91 cm) of 21st-century rise. Assuming the naive forecast has predictive skill over roughly half the calibration window, the extrapolation of the 32-year satellite record provides predictive skill over the next 15–20 years. Beyond that horizon, the naive forecast is less reliable but nevertheless establishes a scenario-independent, data-driven reference (Fig. 9a).

The next class of data-driven forecasts are the semi-empirical models that fit the observed

sensitivity of the rate of GMSLR to changes in GMST (Fig. 7b). Under current warming trends, Rahmstorf (2007) (*Rahmstorf*, 2007) predicts 0.67 m (0.63–0.71 m) of 21st-century rise, matching the naive forecast and the results of this study within error.

At intermediate-scale complexity sit the component level models *** Our work complements and is differentiated from previous studies (*Kopp et al.*, 2014; *Mengel et al.*, 2016; *Wong et al.*, 2017) by the models used for the individual components, the data used in the calibration, the approach to estimating uncertainties, and the anchoring of the projections in the near-term to the observed trends. ***Briefly describe existing models***

Here, we construct a data-driven sea-level projection framework that treats individual sea-level contributors with their own parsimonious physical models calibrated by independent observational records using Bayesian inference. We name the framework the Sea-level Projections from Empirical Calibration of Thermodynamical, Reduced-physics Models (SPECTRUM). The individual components include ocean thermal expansion (thermosteric), Greenland Ice Sheet (GrIS) surface mass balance (SMB) and glacier discharge, West Antarctic Ice Sheet (WAIS) glacier discharge, East Antarctic Ice Sheet (EAIS) SMB, Antarctic Peninsula total ice loss, glaciers (land ice not in the ice sheets), and terrestrial water storage. Some of the component models are adapted from the literature. In these cases, the main contributions from this study are in the data calibration, validation, model synthesis, and forecasting approaches. A notable exception is Greenland Ice Sheet discharge, whose 3-parameter model connecting glacier discharge and ocean temperature is developed in this study. As an overview on validation: Observed GMSLR and its sensitivity to observed global mean surface temperature (GMST) are withheld from the component-level calibrations and serve as diagnostics for the composite model, in which the individual components are added. Unless otherwise noted, all sea-level values in the main text are global-mean 21st-century contributions referenced to the year 2000. Throughout, emission scenarios are labeled by their approximate median end-of-century global mean surface temperature rounded to the nearest integer and referenced to the 1850-1900 (pre-industrial) mean: 2 °C (SSP1-2.6), 3 °C (SSP2-4.5), and 4 °C (SSP3-7.0).

2 A data-driven, component-level projection framework

2.1 Component models, calibration, and validation

2.1.1 Greenland Ice Sheet

The Greenland Ice Sheet (GrIS) is the largest land-ice contributor and has lost mass every year since the 1990s (*Mouginot et al.*, 2019; *Mankoff et al.*, 2021). That loss is the imbalance between solid-ice discharge and surface mass balance (SMB), where SMB is the difference between mass gained by snow accumulation and mass lost to melting.

GrIS discharge is sensitive to ocean temperature and surface melt runoff (*Straneo et al.*, 2013; *Slater et al.*, 2020; *Wood et al.*, 2021). Warm, subsurface waters thin and undercut the calving fronts of marine-terminating glaciers, driving increased calving and net terminus retreat. This effect is amplified where surface runoff drives mixing and buoyant upwelling along the calving face (*Slater and Straneo*, 2022). Surface melt also lubricates glacier beds, but that effect is generally short-lived and localized as the subglacial hydrological system

evolves to accommodate changes in meltwater flux (*Schoof, 2010; Tedstone et al., 2013; Ultee et al., 2022*).

Focusing on the strong influence of ocean-induced melting on discharge, we approximate the GrIS-wide discharge as

$$D = \gamma T_{\text{ocean}}(t - \delta) + r_0, \quad (1)$$

where D is the GrIS discharge in sea-level equivalent units, γ is an effective discharge sensitivity to ocean temperature, δ is the lag between distal ocean warming and ice sheet response, r_0 is the background discharge rate at the baseline ocean temperature, and T_{ocean} is the ocean temperature anomaly in the ocean waters around Greenland relative to the 1995–2005 mean. Because D is the gross discharge flux rather than an anomaly, it becomes a contribution to GMSL only when combined with SMB, whose accumulation term offsets most of the background rate r_0 . We calibrate against total GrIS discharge, integrated across the 219 marine-terminating outlet glaciers resolved in the observational record (*Mouginot et al., 2019*) to trade spatial resolution for signal-to-noise in the data. Our goal is to capture ice-sheet scale discharge rather than the individual basins and glaciers.

The calibration is carried out on the time-integrated form of Eq. 1, which gives the cumulative contribution of GrIS discharge to GMSLR,

$$H_D(t) = \int_{t_0}^t D dt', \quad (2)$$

with γ and r_0 set constant in time. Fitting the time-integrated (cumulative) record further improves the signal-to-noise ratio, because the cumulative signal grows linearly in time while the annual errors accumulate only as its square root. It also directly constrains the total mass lost, the quantity of interest in this study.

The calibration uses 47 years of glacier discharge data (*Mouginot et al., 2019*) and ocean temperatures from EN4 (v4.2.2, g10 bias correction) (*Good et al., 2013*), averaged over 200–500 m depth, the range most relevant to Greenland’s tidewater grounding lines (*Slater et al., 2020*), in the seas surrounding Greenland south of 80°N (58–80°N, 75–5°W), a limit set by the sparse subsurface sampling farther north (Fig. 2). Because the data are resolved annually, the lag δ is in integer-year values. We therefore fit (γ, r_0) by weighted least squares separately for a range of δ values, using the demeaned cumulative record. Each fit carries a penalty on the mismatch between the modeled discharge rate at the end of the record and the rate observed over the final decade, which keeps the calibration consistent with the present-day discharge rate. The posterior is then formed as a Gaussian mixture of the per- δ fits, with weights derived from the Bayesian Information Criterion. This procedure places 99.5% of the weight on $\delta = 5$ yr and gives posterior medians $\gamma = 0.365 \text{ mm SLE yr}^{-1} \text{ }^\circ\text{C}^{-1}$ (0.312–0.419 mm SLE yr^{−1} °C^{−1}) and $r_0 = 1.383 \text{ mm SLE yr}^{-1}$ (1.367–1.399 mm SLE yr^{−1}). The calibration target is a cumulative record, resulting in correlated, so we report parameter intervals from a covariance that accounts for that correlation and for the excess scatter of the residuals relative to the reported observational errors (reduced chi-square $\chi^2/\text{dof} = 1.8$, where χ^2 is the error-weighted sum of squared residuals and $\text{dof} = 45$ is the number of annual observations less the two fitted parameters), rather than from the least-squares covariance, which assumes independent errors. We validate the model against a discharge record spanning 1986–2023 that was withheld from calibration (*Mankoff et al., 2021*). This is

a weak validation, because the *Mouginot et al. (2019)* and *Mankoff et al. (2021)* records share the same underlying raw data, though their independent processing still makes agreement meaningful. Our model falls within error of validation data, as does the total Greenland contribution to GMSLR (Fig. 1c). The validation results suggest that our simple model, with two fitted parameters and a selected lag, captures the total contribution of GrIS discharge to GMSLR during the observational record and its sensitivity to ocean temperatures in Greenland’s peripheral waters.

The fitted $\delta = 5$ yr is an effective lag for spatially averaged forcing and we therefore do not attempt to attribute it to a single process. It lies within the range of timescales spanned by the ocean-to-ice pathway, from exchange between the continental shelf and fjord interiors through the terminus response to undercutting and the inland transmission of retreat to the flux gates at which discharge is measured (*Straneo and Heimbach, 2013; Wood et al., 2021; Robel et al., 2018*). Because T_{ocean} is averaged over Greenland’s peripheral waters, δ also absorbs the time required for a warming anomaly to become representative of conditions at outlet glaciers distributed around the ice sheet; retreat began in southeast and west Greenland in the mid-1990s and reached the northwest a decade later (*Straneo and Heimbach, 2013*).

No published study reports a Greenland-wide sensitivity of discharge rate to ocean temperature, so we derive a simple dimensional scaling to contextualize the inferred value $\gamma = 0.365 \text{ mm SLE yr}^{-1} \text{ }^{\circ}\text{C}^{-1}$. For a single outlet glacier, the submarine melt rate scales approximately linearly with ocean thermal forcing, the ocean temperature above the in situ freezing point T_f . Because T_f is set by the pressure and salinity at the calving face and is effectively constant over the observational record, a change in thermal forcing equals a change in ocean temperature, and the melt-rate perturbation for glacier i is

$$\Delta \dot{m}_i = \kappa_i T_{\text{ocean}}, \quad (3)$$

where κ is the melt sensitivity to thermal forcing averaged over the submerged calving face. Plume parameterizations calibrated against Greenland observations (*Xu et al., 2013; Rignot et al., 2016; Slater and Straneo, 2022*) give a maximum melt sensitivity near the grounding line of $90\text{--}260 \text{ m yr}^{-1} \text{ }^{\circ}\text{C}^{-1}$, and the face-averaged rate is two to three times smaller (*Wood et al., 2021*). We therefore take $\kappa = 36\text{--}104 \text{ m yr}^{-1} \text{ }^{\circ}\text{C}^{-1}$, with a central value of $60 \text{ m yr}^{-1} \text{ }^{\circ}\text{C}^{-1}$. Submarine melt thins and undercuts the terminus, triggering calving and grounding-line retreat, such that the total ice flux perturbation is a factor μ times the direct submarine melt flux. For Greenland’s tidewater glaciers, observations and models suggest $\mu = 3\text{--}10$ (*Enderlin and Howat, 2013; Wood et al., 2021*). The flux perturbation from a single glacier with submerged calving-front area A_f is then

$$\Delta \dot{Q}_i = \mu_i \rho_{\text{ice}} A_{f,i} \kappa_i T_{\text{ocean}}, \quad (4)$$

where $\rho_{\text{ice}} = 900 \text{ kg m}^{-3}$ is the mass density of ice. Summing over Greenland’s $N = 219$ marine-terminating outlets (*Mouginot et al., 2019*) we approximate the scale of γ as

$$\tilde{\gamma} = \frac{1}{A_{\text{ocean}} \rho_{fw} T_{\text{ocean}}} \sum_{i=1}^N \Delta \dot{Q}_i, \quad (5)$$

where $A_{\text{ocean}} = 3.625 \times 10^{14} \text{ m}^2$ is the ocean area and $\rho_{fw} = 1000 \text{ kg m}^{-3}$ is the density of fresh water, which converts added ice mass to sea-level equivalent under the convention $1 \text{ mm SLE} = 362.5 \text{ Gt}$ used throughout this study. Submerged calving-front areas of 0.9–2.7 km^2 are reported for five West Greenland outlets (*Rignot et al.*, 2016). Evaluating Eq. 5 across the full parameter ranges gives

$$\tilde{\gamma} = 0.05\text{--}1.5 \text{ mm SLE yr}^{-1} \text{ }^\circ\text{C}^{-1}, \quad (6)$$

which contains our calibrated $\gamma = 0.365 \text{ mm SLE yr}^{-1} \text{ }^\circ\text{C}^{-1}$. We report Eq. 6 as an order-of-magnitude check on the inferred value. The range spans a factor of 30 because it is the product of four quantities that are not measured in aggregate, and the factorization $\sum_i \mu_i A_{f,i} \kappa_i \approx N \bar{\mu} \bar{A}_f \bar{\kappa}$ assumes these are uncorrelated across outlets, whereas the deepest and widest fronts occupy the warmest fjords. Finally, the calibrated γ is an effective sensitivity that absorbs the covarying atmospheric pathway because increased subglacial runoff amplifies plume-driven melting alongside ocean warming.

2.1.2 Thermosteric

The ocean is the dominant reservoir for the excess heat accumulated by the climate system (*Cheng et al.*, 2017), and the resulting expansion – the thermosteric contribution – is currently the largest single contributor to observed sea level rise (*Frederikse et al.*, 2020). Both heat exchange between ocean and atmosphere and ocean dynamics set the structure of a simple model of the thermosteric effect. Put simply, the ocean warms with a delay relative to the atmosphere, heat enters at the sea surface and is carried downward by turbulent mixing and by the overturning circulation, causing temperature of the interior to reflect the history of surface warming. Downward transfer spans a wide range of timescales. The upper few hundred meters exchange heat with the atmosphere over years to decades, whereas water below the thermocline is renewed over centuries. We therefore represent the ocean as two layers, a fast upper layer coupled to the atmosphere and a slow deep layer coupled only to the upper layer, following the structure of *Geoffroy et al.* (2013). This structure reproduces both the lag between surface warming and expansion and the continued expansion that follows a stabilization of surface temperature, where the deep layer draws heat from the upper layer after the upper layer has equilibrated.

Let us first consider an upper layer of thickness $h_u = 700 \text{ m}$ warmed by $\Delta\theta(z, t)$, which raises steric height of the ocean over time as

$$H_u = \int_0^{h_u} \alpha_{sw} \Delta\theta \, dz, \quad (7)$$

where $\alpha_{sw}(z)$ is the thermal expansion coefficient of seawater, which is depth dependent due to its temperature dependence. The temperature anomaly profile is defined as $\Delta\theta = S_u \varphi$, where $S_u(t)$ is the amplitude of the anomaly in degrees and $\varphi(z)$ is a shape factor. Expanding $\alpha_{sw}(z) = \alpha_{0sw}(z) + \alpha'_{sw} \Delta\theta$, with $\alpha'_{sw} = \partial\alpha_{sw}/\partial\theta$, we rewrite Eq. 7 as

$$H_u = a_u S_u^2 + b_u S_u + \epsilon_u(t - t_0), \quad (8)$$

where $a_u = \int_0^{h_u} \alpha'_{sw} \varphi^2 dz$ and $b_u = \int_0^{h_u} \alpha_{0sw} \varphi dz$. The linear term b_u is the expansion of the upper layer for a given warming while the quadratic term a_u comes from the temperature

dependence of α_{sw} . We added $\epsilon_u(t - t_0)$ to absorb instrument drift during calibration. As GMST, $T(t)$, changes, the amplitude evolves as

$$\frac{dS_u}{dt} = \frac{T - S_u}{\tau_u}, \quad (9)$$

where τ_u is the characteristic time over which the thermal state S_u relaxes to GMST. Equation 9 shows that S_u is an average of past temperatures, weighted toward the recent past over a time window set by τ_u . The upper layer therefore smooths year-to-year fluctuations in GMST and follows longer trends with a delay comparable to τ_u .

The deep layer follows the same construction, with $\Delta\theta_d(z, t) = S_d(t) \varphi_d(z)$ over 700–2000 m depth and $b_d = \int_{h_u}^{h_d} \alpha_{0sw} \varphi_d dz$, with $h_d = 2000$ m. We drop the quadratic term because S_d remains small enough that the temperature dependence of α_{sw} can be neglected. Warming from GMST reaches the deep layer only after passing through the upper layer, so the thermal state S_d relaxes toward S_u as

$$\frac{dS_d}{dt} = \frac{S_u - S_d}{\tau_d}, \quad (10)$$

where we fixed $\tau_d = 150$ yr at the CMIP5 multi-model mean deep-ocean adjustment time (*Geoffroy et al.*, 2013, Table 4), because the 21-year 0–2000 m data record is too short to constrain a centennial relaxation time. Adding this layer gives the total steric response for the upper 2000 m as,

$$H_{ud}(t) = H_u(t) + b_d S_d, \quad (11)$$

which we fit jointly to the 0–700 m and 0–2000 m records. Finally, to account for expansion below 2000 m depth, we define

$$H_{\text{full}}(t) = H_{ud}(t) + \dot{H}_{>2000}(t - t_{\text{ref}}), \quad (12)$$

where $\dot{H}_{>2000} = 0.07 \pm 0.04$ mm yr^{−1} (*Purkey and Johnson*, 2010) is a constant rate for depths below 2000 m, referenced to $t_{\text{ref}} = 2000$. This layer is not observed at annual resolution, so its rate is adopted rather than calibrated.

We fit the two-layer model in Eq. 11 jointly to NOAA in situ thermosteric sea level at 0–700 m (1955–2025) and 0–2000 m (2005–2025) (*Levitus et al.*, 2012) with Bayesian inference, in which MCMC samples parameter combinations $\phi = (a, b_u, b_d, c, \tau_u, \sigma_{\text{extra}})$ in proportion to their posterior probability (Fig. 3). By Bayes’ theorem, the posterior probability equals the likelihood of the data given ϕ times the prior probability of ϕ . Assuming the residual at each point is normally distributed about the model with variance σ_i^2 , the negative log-likelihood takes the form $\frac{1}{2} \sum_i [(r_i(\phi)/\sigma_i)^2 + \ln \sigma_i^2]$. Adding the negative log-prior for τ_u , whose log-normal prior (defined below) has mean μ_{τ_u} and standard deviation s_{τ_u} in $\ln \tau_u$ (the other five parameters have flat priors and so contribute only an additive constant, which we drop), gives the objective function

$$\mathcal{J}(\phi) = \sum_i \left[\left(\frac{r_i(\phi)}{\sigma_i} \right)^2 + \ln \sigma_i^2 \right] + \left(\frac{\ln \tau_u - \mu_{\tau_u}}{s_{\tau_u}} \right)^2, \quad (13)$$

which is twice the negative log-posterior up to an additive constant, so MCMC visits parameter combinations ϕ with probability proportional to $\exp[-\mathcal{O}(\phi)/2]$, favoring combinations

that fit the data well (small $\mathcal{O}$) over those that fit poorly (large $\mathcal{O}$). Here $r_i(\phi)$ is the residual between the demeaned model and demeaned observation at point i , and σ_i^2 is that point’s total variance. The point variance is defined by the quadrature sum of the formal measurement error, a structural uncertainty retained after a bias correction in the NOAA thermosteric observations (*Li et al.*, 2022), and σ_{extra} , a noise term, shared across both the 0–700 m and 0–2000 m records, that absorbs misfit beyond the reported observational error. The MCMC sampling uses package *emcee* with 64 walkers, 6000 burn-in steps, and 30,000 production steps to ensure convergence. The relaxation time of the upper layer, τ_u , has an informative prior, $\log \tau_u \sim \mathcal{N}(\ln 8, 0.5^2)$, or $\mu_{\tau_u} = \ln(8 \text{ yr})$ and $s_{\tau_u} = 0.5 \text{ yr}$, consistent with the range in *Geoffroy et al.* (2013). b_u and τ_u are correlated in the likelihood—a slower upper-layer response can be partly offset by a larger sensitivity to fit the same trajectory—but this partial degeneracy is managed by the prior on τ_u . Five parameters — a , b_u , b_d , c , and σ_{extra} — get uninformative flat priors with wide bounds (Fig. 3). Both records are demeaned before fitting. The posterior median and 90% credible intervals are: $a = 20.0$ [10.1, 31.2] mm $^{\circ}\text{C}^{-2}$, $b_u = 51.0$ [40.6, 75.4] mm $^{\circ}\text{C}^{-1}$, $b_d = 68.6$ [37.0, 104.3] mm $^{\circ}\text{C}^{-1}$, $c = 0.023$ [−0.112, 0.140] mm yr^{-1} , $\tau_u = 11.6$ [6.2, 25.3] yr, and $\sigma_{\text{extra}} = 2.0$ [1.7, 2.3] mm. The calibrated model reproduces both records with RMS residuals of 2.3 mm for depths 0–700 m and 1.3 mm for 0–2000 m, against 42 and 30 mm of observed rise over the respective records.

We validate the full-depth model in Eq. 12 against the independently derived results of *Frederikse et al.* (2020), which are withheld from calibration (Fig. 4c,f). *Frederikse et al.* (2020) reconstruct global-mean steric sea level for 1900–2018 by combining in situ subsurface ocean-temperature products over 1957–2018 with a reconstruction of historical ocean heat storage for the pre-1957 period. Our comparison uses their 1957–2018 portion, showing good agreement with a relative RMS of 2.5 mm (Fig. 4c,f). For completeness, we also show a pre-1957 comparison, where our model and the reconstruction broadly agree, with an RMS of 3.8 mm.

2.1.3 Glaciers

The rate of glacier mass loss has been shown to be approximately linear with warming (*Rounce et al.*, 2023; *Zemp et al.*, 2025). In the interest of developing consistent uncertainty quantification for this study, we fit cumulative glacier sea-level contribution in level space as

$$H_{\text{gl}}(t) = b_{\text{gl}} \int_{t_0}^t T d\tau + c_{\text{gl}}(t - t_0) + H_{\text{gl}0} + \varepsilon_{\text{gl}}(t), \quad (14)$$

where T is the GMST anomaly, the integrals are computed on a monthly grid and evaluated at annual observation times, $H_{\text{gl}0}$ is a baseline offset, and $\varepsilon_{\text{gl}}(t) \sim \mathcal{N}(0, \sigma_{\text{gl}0\text{bs}}^2 + \sigma_{\text{gl}0\text{extra}}^2)$ with $\sigma_{\text{gl}0\text{extra}}$ an additional model-inadequacy noise term learned from the data. We fit the model to the demeaned Glacier Mass Balance Intercomparison Exercise (GlaMBIE) record (2000–2023, 23 annual points) using Bayesian inference with weakly informative priors $b_{\text{gl}} \sim \text{HalfNormal}(\sigma = 0.087 \text{ mm yr}^{-1} ^{\circ}\text{C})$ encompassing previously published values and $c_{\text{gl}} \sim \mathcal{N}(0.3, 1) \text{ mm yr}^{-1}$ (*Rounce et al.*, 2023; *Zemp et al.*, 2025). For the MCMC, we use package *emcee* with 32 walkers, 2000 burn-in steps, and 4000 production steps. The posterior median sensitivity is $b = 0.66 \text{ mm yr}^{-1} ^{\circ}\text{C}^{-1}$ [0.58–0.75] and the committed rate is $c = 0.57 \text{ mm}$

yr^{-1} [0.51–0.63], with intervals computed from a covariance that accounts for the correlated errors of this cumulative calibration record (Fig. 5a).

2.1.4 Antarctic Peninsula

The Antarctic Peninsula component uses the same level-space Bayesian framework as with glaciers, with the rate–temperature model $\dot{H}_p = a_p T^2 + b_p T + c_p$ calibrated against the Ice Sheet Mass Balance Intercomparison Exercise (IMBIE) Peninsula record (1992–2020, 29 annual points; *Otosaka et al.*, 2023) via emcee (32 walkers, 2000 burn-in steps, and 4000 production steps). We fit both the linear ($a_p = 0$) and quadratic model. The Bayesian Information Criterion favors the linear model with posterior median sensitivity $b_p = 0.055 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-1}$ [−0.031–0.138] and committed rate $c_p = 0.028 \text{ mm yr}^{-1}$ [0.003–0.052] (Fig. 5b). The Peninsula contribution reflects combined SMB and glacier discharge responses to warming.

2.1.5 East Antarctica

The EAIS component follows the same framework as the Peninsula and is fit to the IMBIE EAIS record (1992–2020, 29 annual points; *Otosaka et al.*, 2023). The linear sensitivity b_e is given a symmetric prior, $b_e \sim \mathcal{N}(0, \sigma = 0.087 \text{ mm yr}^{-1} \text{ }^\circ\text{C})$. We fit a linear model, with intervals computed from the same correlation-aware covariance as glaciers and the Peninsula. The results give a posterior sensitivity $b_e = -0.030 \text{ mm yr}^{-1} \text{ }^\circ\text{C}^{-1}$ [−0.177–0.118] and committed rate $c_e = -0.023 \text{ mm yr}^{-1}$ [−0.084–0.040] (Fig. 5c).

2.1.6 West Antarctica

Unlike glaciers, the Peninsula, and EAIS, we do not fit a rate–temperature regression to WAIS over the observational record. We instead use the IMBIE observations directly in the budget closure (§2.2), setting the WAIS contribution to the total component-wise sum equal to the observed cumulative record itself, with its reported measurement uncertainty (Fig. 5d). We set the rate of WAIS loss to zero prior to 1992, when the IMBIE begins and the fit to the IMBIE record shows a rate $< 0.05 \text{ mm/yr}$.

2.2 Total GMSL budget closure (validation)

Satellite altimetry data collected from 1993–2025 provide observations of GMSL (*Beckley et al.*, 2017; *Hamlington et al.*, 2020) that can be compared with the sum of the independently calibrated model components (Fig. 6). Total GMSL is withheld from calibration to serve as a validation data set. For Greenland SMB, which we do not calibrate due to data sparseness, we use RACMO-derived SMB (*Mouginot et al.*, 2019). Terrestrial water storage is adopted from *Frederikse et al.* (2020).

The component sum closes 93–101% of the observed GMSLR rate over three epochs (1993–2025, 2000–2025, 2010–2025), offering validation that the individually calibrated components together approximate GMSL during the instrumental record. Because each component is calibrated independently, we treat their rate uncertainties as independent and combine them in quadrature. Each component’s individual uncertainty comes directly from

its Monte Carlo ensemble. For a given epoch, we fit a linear trend to every ensemble member and take the standard deviation of the resulting trends. Repeating this for each epoch lets the combined uncertainty differ from one epoch to the next, rather than using one fixed value throughout the satellite record. By this measure, the residual is within one combined standard deviation of the component rate uncertainties in every epoch (0.72σ , 0.09σ , and 0.60σ for 1993–2025, 2000–2025, and 2010–2025, respectively).

2.3 Forecasting methods and validation

The forecasts presented here use the IPCC AR6 GMST trajectories for different emission scenarios, though they can be applied to any GMST trajectory. We focus on the policy-relevant emission scenarios, which are labeled by their approximate median end-of-century global mean surface temperature anomaly (relative to preindustrial) rounded to the nearest integer: 2°C (SSP1-2.6), 3°C (SSP2-4.5), and 4°C (SSP3-7.0). We initialize the GMSL forecast using the GMSL level, rate, and acceleration observed in the satellite-era and blend in the component-model sum in rate space, such that

$$\dot{H}_{\text{forecast}} = w\dot{H}_{\text{obs}} + [1 - w]\dot{H}_{\text{comp}}, \quad (15)$$

where $\dot{H}_{\text{obs}}$ is the rate from the satellite-era quadratic fit of *Hamlington et al.* (2024), $\dot{H}_{\text{comp}}$ is the rate of the sum of the calibrated components. The blending weight $w(t)$ is a sigmoid

$$w = 1 - \Theta\left(\frac{t - t_r}{\tau_r}\right), \quad (16)$$

where $\Theta(x) = (1 + e^{-x})^{-1}$ is the standard logistic function, $t_r = 2035$, and $\tau_r = 5$ yr. At $t \ll t_r$, $w \approx 1$ and the forecast follows the observed quadratic trajectory. At $t \gg t_r$, $w \approx 0$ and the forecast follows the component-model physics. The transition spans approximately 22 yr, meaning that 10%–90% of the weight shift occurs over $\pm 2.2\tau_r \approx \pm 11$ yr around t_r . We chose $t_r = 2035$ because it is more than a quarter and less than a half of the GMSL satellite altimetry record into the future, a period when the extrapolation of observations should have strong forecasting skill. The tapering recognizes the expected reduction in forecasting skill of the simple data extrapolation over timescales longer than roughly half of the observational window. Pushing t_r to 5 years later (approximately half of the calibration window and arguably the latest viable choice) produces forecasts that are indistinguishable within uncertainty, making the results insensitive to the specific value of t_r within the range of 10–15 years into the future.

The GMSL forecast is obtained by integrating Eq. (15) forward from the observed level at the forecast origin ($t_0 = 2025$, the most recent NASA altimetry data as of this writing), so that

$$H_{\text{forecast}}(t) = H_{\text{obs}}(t_0) + \int_{t_0}^t \dot{H}_{\text{forecast}}(t') dt'. \quad (17)$$

The blending is done sample-by-sample across the 2000-member Monte Carlo ensemble, preserving correlations between the quadratic-fit parameter uncertainty and the component-model uncertainty. Because the extrapolation is derived entirely from the satellite altimetry

record, it carries no dependence on emissions pathway. The component sum, $\dot{H}_{\text{comp}}$, is SSP-dependent through the temperature-forced component models. Consequently, the blended forecast is SSP-independent in the near term (where $w \approx 1$) and becomes progressively SSP-dependent as the component physics dominate at later times.

Uncertainty in $\dot{H}_{\text{comp}}$ is propagated by Monte Carlo summation of the components. Each component has its own 2000-member ensemble of level-space projections $H_k(t)$, drawn independently from that component’s posterior or, for components using prescribed literature sensitivities rather than data calibration, from the corresponding parameter distributions. The total level is formed sample-by-sample, $H_{\text{comp},i}(t) = \sum_k H_{k,i}(t)$ for $i = 1, \dots, 2000$, so draws from independently calibrated components are summed without pairing individual samples by any shared physical driver. The component rate $\dot{H}_{\text{comp}}$ used in Eq. (15) is obtained by finite-differencing this summed ensemble in time before blending with the observational quadratic rate. Reported credible intervals (66% and 90%) are the 17th/83rd and 5th/95th empirical percentiles of the resulting ensemble at each time step.

2.3.1 Component-level GMST dependence

Not all of the SPECTRUM model components described in Section 2.1 have direct, explicit dependences on GMST. These components include the Greenland glacier discharge, which depends on the temperature of the ocean surrounding Greenland, and the SMB of Greenland and EAIS.

To project GrIS discharge model forward in time, we calibrate a transfer function between the Greenland regional surface temperature anomaly and the ocean temperature and then apply a normal distribution of Arctic amplification factors to calculate the Greenland regional surface temperature from projected GMST under different emission scenarios. The linear transfer function on annual-mean observations is

$$T_{\text{ocean}}(t) = \xi T_{\text{surface}}^{\text{Gr}}(t) + \beta + \varepsilon_{go}(t), \quad (18)$$

where $T_{\text{surface}}^{\text{Gr}}$ is the Greenland regional surface temperature anomaly and ε_{go} is a zero-mean residual. Both series are anomalies relative to the 1995–2005 baseline. An OLS fit over 52 common annual values gives $\xi = 0.148 \pm 0.027$, $\beta = -0.051 \pm 0.031$ °C, and residual standard deviation $\sigma_{\varepsilon_{go}} = 0.20$ °C.

For projections, SSP GMST trajectories are converted to Greenland regional surface temperature using an Arctic amplification factor $AA = 3.0 \pm 0.5$, consistent with the observed ratio of Greenland-to-global warming over 1970–2021. Greenland regional surface temperatures are then passed through Eq. (18) to obtain projected T_{ocean} , which then enters Eq. (2) via the delayed integral. Each of the 2,000 Monte Carlo samples draws $AA \sim \mathcal{N}(3.0, 0.5^2)$, $\xi \sim \mathcal{N}(0.148, 0.027^2)$, and $\beta \sim \mathcal{N}(-0.051, 0.031^2)$ jointly with (γ, r_0, δ) from the discharge posterior, so that Arctic amplification, transfer-function, and discharge parameter uncertainties all propagate into the projections. A single draw of $(\alpha, \beta, \gamma, r_0, \delta)$ is applied to all SSP scenarios, so that inter-SSP differences isolate the effect of the emissions pathway from parameter uncertainty.

For SMB in Greenland and EAIS, the forecast uses simple thermodynamic models calibrated from regional climate models. For Greenland SMB, the forecast uses a quadratic temperature scaling $\dot{H}_{\text{smb}} = C_{T^2} \Delta T^2 + C_T \Delta T + \dot{M}_0$ with sensitivities drawn from the spread

between regional climate models (*Fettweis et al.*, 2020; *Noël et al.*, 2021; *Glaude et al.*, 2024). The linear sensitivity $C_T \sim \mathcal{N}(-300, 80^2)$ Gt yr⁻¹ °C⁻¹ of GMST encompasses the MAR, RACMO, and HIRHAM estimates. The quadratic sensitivity, $C_{T^2} \sim \mathcal{N}(-50, 30^2)$ Gt yr⁻¹ °C⁻², represents the effects of ablation zone expansion and melt-albedo feedback with warming. $\dot{M}_0 = 380$ Gt yr⁻¹ is the baseline SMB. For EAIS SMB, we use a linear model ($C_{T^2} = 0$) and the literature-derived accumulation sensitivity $C_T = 60 \pm 20$ Gt yr⁻¹ °C⁻¹ of GMST (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013), reflecting the Clausius–Clapeyron scaling of snowfall with warming ($C_T > 0$ corresponds to mass gain, i.e., negative SLR). Data from IMBIE show EAIS in approximate balance (Fig. 3), so we set $\dot{M}_0 = 0$. The EAIS contribution is small (−13 to −30 mm at 2100, depending on scenario) relative to the total, meaning the GMSL projection results of this study are largely insensitive to these parameter choices.

Terrestrial water storage projections are adopted directly from the IPCC AR6 rather than independently modeled, as the processes governing groundwater depletion, reservoir impoundment, and land hydrology changes are outside the scope of this framework and the data are insufficient to allow for a data-driven approach as with other components. At each decadal interval in the IPCC projections, we fit a Gaussian distribution to the reported median and inter-quantile spread and draw 2,000 Monte Carlo samples, which are then linearly interpolated to annual resolution. For the hindcast period (pre-2020), we use the *Frederikse et al.* (2020) TWS observations with Monte Carlo sampling from their reported uncertainties. TWS is SSP-dependent but the contribution is small (~ 30 mm at 2100 under SSP2-4.5) and does not materially affect the total uncertainty budget.

2.3.2 Validation: Hindcasting the level, rate, and transient sensitivity

We validate the combined GMST-sensitive model by two hindcasting approaches. Both are generated by forcing the model components with observed GMST back to 1900 (*Rohde and Hausfather*, 2020), with an emphasis on post-1950 when increasingly reliable GMSL data are available. The first comparison is between global mean sea level and rate produced by SPECTRUM with data from *Fox-Kemper et al.* (2021) and *Frederikse et al.* (2020). For the second comparison, we calculate the transient sensitivity of the SPECTRUM output and GMSL data, where the transient sensitivity is defined as $\alpha = d\dot{S}/dT$ with $\dot{S}$ as the rate of change in GMSL. Here, we use the same t_0 starting point (2025, the end of the satellite altimetry record) and tapering scheme as described in the forecasting section (§2.3) to ensure the hindcast differs only from the forecast in the GMST trend used to drive the model.

Extending the hindcast back to 1900 shows two distinct regimes relative to the GMSL reconstructions (Fig. 7a). From approximately 1950 to the present, the hindcast agrees closely with the *Frederikse et al.* (2020) and (*Fox-Kemper et al.*, 2021) GMSL reconstructions and with the satellite altimetry record (*Beckley et al.*, 2017). The hindcast rate of GMSL rise over 1950–2000 (1.11 mm yr⁻¹) matches both reconstructions (1.02 and 1.01 mm yr⁻¹), and the hindcast level at 1950 is within 10 mm of each. Before 1950, the hindcast diverges from the *Frederikse et al.* (2020) reconstruction, understating the reconstructed rate of rise (0.20 mm yr⁻¹ versus 1.39–1.64 mm yr⁻¹ over 1900–1950) and level (roughly 70–75 mm low by 1900, relative to the 2000 baseline). This divergence coincides with the period in which the reconstructions rely increasingly on sparse tide-gauge coverage and model-based infilling and our model suffers from a lack of constraints on mass changes in the Greenland Ice Sheet

and glaciers. The limitations of our model arise by design: we chose to focus only on model representations that can be fit to observations and that represent the ongoing response to warming. This choice excludes more complete and complex land-ice process representations that would be necessary to represent GMSL during colder periods in the past, as this is not our goal. We therefore treat the pre-1950 hindcast as qualitatively informative but do not draw quantitative conclusions from it. We restrict our validation claims to the 75-year, post-1950 period, where component calibration data are available and the agreement with observations is close.

The transient sensitivity of our component model, which defines the change in the rate of change in GMSL with warming, tracks closely with the observed sensitivities (Fig. 7b). It is worth emphasizing that this agreement includes the transition from the earlier period when thermosteric sea-level rise of the largest contributor to GMSL to the present when the sum of land-ice and terrestrial water storage (barystatic) are larger overall contributors to GMSL. The transient sensitivity of the observations and model also agree with the semi-empirical results of *Rahmstorf* (2007) (Fig. 7b), which continue to track observations nearly 20 years after publication, showing that the simple semi-empirical model continues to provide a reasonable first-estimate of GMSL rise with warming.

Together, the level, rate, and transient-sensitivity comparisons indicate that our framework reproduces the observed behavior of GMSLR over a 75-year window (1950–2025) spanning roughly 1 °C of GMST change, without having been calibrated on observations of GMSL itself. This window includes three forms of agreement: the hindcast level tracks observations to within 10 mm, the hindcast rate matches observed decadal trends to within about 10%, and the hindcast transient sensitivity reproduces both the observed rate–temperature curvature and its underlying physical cause (the shift from thermosteric to barystatic dominance). To our knowledge, no other GMSL forecasting framework or model presents this level of detailed validation nor hindcast skill. Because this validation window is comparable in duration and magnitude of warming to the near- and mid-term forecast horizon, it supports extending the same confidence forward to a similar span, roughly a century and a further 1–2 °C of warming, consistent with the 75-year window and 10-year taper used to blend observed and component-model trends in the forecast itself (§2.3). This validated envelope does not, however, extend to processes for which comparable historical constraints do not exist, which is why WAIS, the largest source of forecast uncertainty, is treated separately through a scenario-based ‘storyline’, rather than data-driven approach, in this work.

2.3.3 West Antarctic Ice Sheet

WAIS is the largest source of uncertainty in sea-level projections (*Fox-Kemper et al.*, 2021). The ice sheet contains ~ 3.3 m of sea level equivalent and rests on land that is below sea level and deepens inland, making it susceptible to internal mass-loss (positive) feedbacks commonly called the marine ice sheet instability (MISI; *Schoof* (2007); *Joughin et al.* (2014)). The delivery of warm, deep seawater is the primary climate driver of mass loss (*Fricker et al.*, 2025). Ocean thermal forcing sufficient to drive retreat is likely committed throughout the 21st century regardless of emissions pathway (*Naughten et al.*, 2023; *Rosier et al.*, 2025) and may be driven as much by ocean processes local to the ice sheet as those further afield (*Moorman et al.*, 2023). Based on these lines of evidence, we make our WAIS projec-

tions independent of the warming trajectory, meaning the range of 21st-century outcomes is controlled by ice dynamics that, to leading order, are insensitive to emissions over our forecasting horizon.

No physical process models have been proven to reproduce observed trends (*Aschwanden et al.*, 2021) and we find insufficient data coverage and duration to support a data-driven forecast like those described above. To provide a representative range of uncertainties, we construct a scenario-mixture framework, or ‘storyline’, meant to broadly represent the scientific literature (Fig.***). This approach cannot provide, nor should it be interpreted as, a robust probability distribution. We define two scenarios entitled ‘slow WAIS’ and ‘fast WAIS’ that are comparable to the medium and low confidence scenarios of IPCC AR6, respectively.

Scenario 1: ‘slow WAIS’, where no marine ice sheet instability operates and mass loss simply continues the observed 1992–2020 IMBIE discharge record and its recent acceleration forward, sampled directly from a Bayesian quadratic-in-time fit to that record rather than a forward physical model. This scenario gives a median of 90 mm and a 90% CI of 61–118 mm GMSLR contribution by 2100. Based on the changes in WAIS discharge over the past decades and the position by the IPCC AR6, we take this scenario to be unlikely, and assign it a 10% probability. Scenario 2, ‘fast WAIS’, represents MISI-driven acceleration and grounding-line retreat, with or without the additional cascading structural failure of ice cliffs known as marine ice cliff instability (MICI), yielding a GMSL contribution of 0.13–1.0 m by 2100. The lower value is pinned to the 99th percentile of Scenario 1’s endpoint distribution, since MISI-triggered outcomes are expected to exceed anything a continued no-instability trend can produce; this coincides closely with an independent basin-by-basin estimate built from transient-calibrated ice sheet model projections (*Goldberg et al.*, 2026) and the observed 33 km grounding-line retreat at Pine Island Glacier since 1992 (*Rignot et al.*, 2026). The upper bound is a round 1 m, chosen so the full two-scenario mixture’s own 95th percentile (1.27 m) sits at or below the IPCC AR6 low-confidence Antarctic Ice Sheet 95th percentile under SSP5-8.5 (*Fox-Kemper et al.*, 2021), rather than pinning Scenario 2’s own percentile to that value directly, which would push the mixture’s tail meaningfully above AR6 low confidence with no additional data to defend the excess; 1 m remains consistent with independent cross-checks (Supplementary Materials). We assign Scenario 2 a 90% probability, consistent with IPCC AR6 ‘deep uncertainty’ framing for the processes it spans. The Scenario 2 endpoint is drawn from a skew-normal distribution in log-space with physically motivated asymmetry (*Robel et al.*, 2019), and a multiplicative rheology correction (R with median $R = 1.3$) is applied to it, reflecting the underestimation of ice loss when models assume the ice-viscosity stress exponent $n = 3$ rather than the observationally constrained value $n = 4.1 \pm 0.4$ (*Millstein et al.*, 2022; *Getraer and Morlighem*, 2025; *Martin et al.*, 2026); this correction is not applied to Scenario 1, which is grounded in observed mass balance rather than an ISMIP6-class ice-sheet model. The resulting mixture has a median 21st century sea-level rise of 0.35 m with a 0.09 to 1.27 m 90% CI at 2100. We refer to this mixed model as ‘fast WAIS’ because it allows for, but does not force, rapid ice mass loss (Scenario 2).

Sensitivity tests of the scenario weight show the chosen weight does not qualitatively impact the findings of this study.

3 Results

We drive the calibrated model components with the IPCC AR6 GMST trajectories of 2 °C (SSP1-2.6) through 4 °C (SSP3-7.0), the pathways most consistent with current policies and stated pledges.

For the 2 °C target, our forecast shows a median of 0.74 m of sea-level rise [0.48–1.60 m] relative to year 2000, which is 70% above the IPCC AR6 medium-confidence estimate of 0.44 m. Under 3 °C warming, approximately the current trajectory, the median is 0.91 m [0.65–1.78 m], 64% above IPCC’s 0.56 m. For the 4 °C trajectory, the median reaches 1.12 m [0.84–1.98 m]. For reference, the 5 °C warming trajectory (SSP5-8.5) yields a median of 1.31 m [1.01–2.16 m].

Relative to pre-industrial GMSL (adding ~ 0.19 m of observed rise through 2000), these forecasts correspond to medians of 0.93–1.31 m of global mean sea level rise across the 2-4 °C policy-relevant pathways. The probability of exceeding 1 m relative to pre-industrial by 2100 is 40% under 2 °C warming, 71% under 3 °C, and 98% under 4 °C warming trajectory. The probability of exceeding 2 m relative to pre-industrial by 2100 is approximately 5%, with a minimum of 3% if warming is held below 2 °C. Under the condition that WAIS loses ice slowly, the probability of exceeding 1 m of sea level rise relative to pre-industrial levels effectively drops to zero for less than 3 °C of warming, implying lower flood risks for tens of millions of people and reduced costs of adaptation and damages of order trillions of US dollars (Fig. 9b). As validation, our component forecast for 3 °C of warming closely agreement with extrapolations of observed GMSLR trends, as should be expected if the warming trend remains similar to the current trend and WAIS, which has contributed only $\sim 4\%$ to GMSLR to date, continues to lose mass at its current rate.

Across the range of possible WAIS mass loss and emission scenarios, our forecasts suggest that WAIS is likely to become the dominant source of sea-level rise before the end of the century (Fig. 10a). It is uncertain when that will happen and depends on how quickly emissions decrease. The stack of mean values shows the relative contributions of the primary sea-level contributors, showing that the thermosteric component likely will remain the single largest contributor in the 21st century unless WAIS mass losses over take it. Greenland and mountain glaciers will continue to lose mass and contribute to sea level rise throughout the 21st century.

The distribution of the uncertainties shows a dramatic separation between the predictable and, currently, unpredictable components (Fig. 10b). These results show that WAIS accounts for virtually all of the variance, and approximately 89% of the confidence interval, in 21st century sea-level rise. The other sea level contributors, all of which have calibrated sensitivities to global mean surface temperature, contribute approximately 1% to the total variance and 11% to the 90% (CI) uncertainty range across the plausible warming scenarios. As a result, the 90% uncertainty range of 1.13 m for 3 °C is three times larger than the spread across warming scenario medians (0.37 m from 2–4 °C). This picture is fundamentally different under the condition that WAIS loses mass slowly. In the slow-WAIS scenario, the warming-sensitive components become the largest contributors to the uncertainty (righthand column of Fig. 10b). The relative magnitude of uncertainties reveals that sea-level risk decomposes into two approximately orthogonal dimensions, each addressed by a distinct class of societal response.

4 Discussion

4.1 Orthogonal dimensions of sea-level risk

The first risk dimension is sensitive to emissions reduction. The temperature-dependent components are sensitive to changes in global mean surface temperature. In this work, we calibrate the sensitivities using decades of observations. Their collective contribution shifts systematically across warming trajectories and is reducible by emissions cuts (Fig. 10a). Under the ‘slow WAIS’ scenario, the forecast narrows to a 90% range of only ~ 14 cm, an order of magnitude narrower than the full scenario cast that allows for rapid mass loss from WAIS due to internal feedbacks (Fig. 9). In the slow scenario, emissions policy is the primary lever for managing risk: the difference between 2°C and 4°C warming is roughly 20 cm of additional rise, which roughly translates to 30 million additional people exposed to coastal flood risks and US\$50B/yr in coastal adaptation costs.

The second risk dimension is relatively insensitive to emissions reductions. WAIS dynamics accounts for virtually all of the total forecast variance and is largely decoupled from the emission pathway. The tail of the forecast distribution—the outcomes that drive the most severe human and economic consequences—is controlled almost entirely by whether WAIS retreat remains gradual at modern rates of loss or transitions to rapid, irreversible retreat. Emissions reduction decreases the median but does not substantively change the tail of the distribution. The probability of exceeding 2 m relative to pre-industrial is 3% under the 2°C trajectory and 8% under 4°C , with a 5% probability across emission scenarios. The 3% lower bound on this range highlights the fact that emission reductions alone cannot retire the high risk outcomes.

The risks and costs to coastal communities helps to highlight the dichotomy between the two possible scenarios of fast vs slow WAIS mass loss, or equivalently, emissions-insensitive vs emissions sensitive (Figs. 9 and 10a). The difference at the 5% exceedance probability, one possible design threshold for coastal adaptation that we denote by stars in Fig. 9b, is 1.1 m of sea-level rise by 2100 under the current policy warming trajectory (3°C). This difference translates into roughly 200M more people exposed to annual flood risks, US\$143B/yr in coastal adaptation, and US\$14.1T/yr ($\sim 12\%$ global GDP per year) in coastal flood damages without further adaptation. These large numbers reflect global population densities, with the highest densities near coastlines, and the fact that the locations of coastal communities and infrastructure are calibrated to lower sea levels. For intuition, the fast-WAIS scenario at 5% exceedance implies an order of magnitude more sea level rise in the next 75 years than was experienced in the past 125 years, while the slow-WAIS scenario implies a factor of 3.5 more rise.

Resolving the question of WAIS mass loss rates is among the most financially consequential goals in modern science (Fig. 10c). A value-of-information analysis shows that perfect knowledge of whether WAIS will lose mass slowly or rapidly is worth of order US\$5T in avoided costs, with the value peaking when confidence in a slow-WAIS future reaches 80% and the information arrives early enough to inform infrastructure decisions. The value decays with time because late-arriving information leaves fewer years of decisions to improve, and because all forecasts, including this study, degrade as they extend beyond their calibration bounds. These numbers frame the research investment question in concrete terms:

even partial reduction of WAIS uncertainty, well short of perfect knowledge, would justify annual research expenditures orders of magnitude above current levels. The return on that investment is determined by how quickly the observations can narrow the range of WAIS outcomes, which depends on both the rate of data collection and the development of reliable process models, like those used in the IPCC, capable of providing explanatory insights that underscore deep understanding of ice sheets and climate.

Our forecast identifies a single measurement that would do more to reduce sea-level uncertainty than any other: the rate of West Antarctic ice loss over the next two decades. If WAIS mass loss continues near current rates through the 2030s and 2040s, that would build the case for the slow-WAIS scenario and allow the forecast to narrow toward the predictable, lower, emissions-sensitive range where adaptation costs are more manageable. If instead grounding-line retreat accelerates beyond the pace of our central scenario, the distribution would shift toward higher outcomes and 1.5–2 m of rise would move from a tail risk to a planning baseline. Either outcome would reshape coastal adaptation worldwide and would justify investment well beyond current levels. The observations needed to distinguish these futures can be made within a generation, but only if the observing systems and research programs are sustained and expanded at a scale that reflects the costs at stake.

Scientific research is a necessary condition for providing actionable information for coastal adaptation, but it is as likely as not that further research will add to the growing evidence to support rapid WAIS mass loss rather than build confidence in a slow-WAIS future (*Fricker et al.*, 2025). Sustaining a wide range of uncertainties that admit < 1 m and > 2 m of GMSLR in 2100 at 10% probabilities, will force decisions on coastal planning, adaptation, and managed retreat that impact millions of lives and cost billions of dollars to be made under large uncertainties. The extent of these uncertainties in terms of lives, property, and ecosystems raises the question: Can we take action to increase the potential stability of West Antarctica? How much harm to people and the environment can we prevent? Recently, some research directions have been proposed to answer this question (*MacAyeal et al.*, 2024) with initial models of glacier interventions aimed at stabilizing WAIS show promise (*Meyer et al.*, 2026). As with any nascent research, there are many unresolved questions about the feasibility, unintended consequences, and governance of these interventions (*Siegert et al.*, 2025). There are risks in intervening to remediate the West Antarctic Ice Sheet and slow its loss and the logistical challenges will undoubtedly be great, but these risks and challenges must be weighed against the risks and challenges of sea-level rise as described in this study. There is no zero risk, zero cost response to sea level rise. The costs and challenges are real and must be borne regardless of what decisions are made. The questions are how much cost, how much challenge, and how much harm. The answers hinge on how fast West Antarctica will lose ice, and nothing short of a virtual guarantee in slow WAIS mass loss provides comfort for the world’s coastlines.

References

Aschwanden, A., T. C. Bartholomäus, D. J. Brinkerhoff, and M. Truffer (2021), Brief communication: A roadmap towards credible projections of ice sheet contribution to sea level, *The Cryosphere*, 15, 5705–5715, doi:10.5194/tc-15-5705-2021.

- Beckley, B. D., P. S. Callahan, I. Hancock, David W., G. T. Mitchum, and R. D. Ray (2017), On the “Cal-Mode” correction to TOPEX satellite altimetry and its effect on the global mean sea level time series, *Journal of Geophysical Research: Oceans*, *122*(11), 8371–8384, doi:10.1002/2017JC013090.
- Cheng, L., K. E. Trenberth, J. Fasullo, T. Boyer, J. Abraham, and J. Zhu (2017), Improved estimates of ocean heat content from 1960 to 2015, *Science Advances*, *3*, e1601545, doi:10.1126/sciadv.1601545.
- Enderlin, E. M., and I. M. Howat (2013), Submarine melt rate estimates for floating termini of Greenland outlet glaciers (2000–2010), *Journal of Glaciology*, *59*(213), 67–75, doi:10.3189/2013JoG12J049.
- Fettweis, X., S. Hofer, U. Krebs-Kanzow, C. Amory, T. Aoki, C. J. Berends, A. Born, J. E. Box, A. Delhasse, K. Fujita, et al. (2020), GrSMBMIP: intercomparison of the modelled 1980–2012 surface mass balance over the Greenland Ice Sheet, *The Cryosphere*, *14*, 3935–3958, doi:10.5194/tc-14-3935-2020.
- Fox-Kemper, B., H. T. Hewitt, C. Xiao, G. Aðalgeirsdóttir, S. S. Drijfhout, T. L. Edwards, N. R. Golledge, M. Hemer, R. E. Kopp, G. Krinner, et al. (2021), Ocean, cryosphere and sea level change, in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, Cambridge University Press, doi:10.1017/9781009157896.011.
- Frederikse, T., F. Landerer, L. Caron, S. Adhikari, D. Parkes, V. W. Humphrey, S. Dangendorf, P. Hogarth, L. Zanna, L. Cheng, and Y.-H. Wu (2020), The causes of sea-level rise since 1900, *Nature*, *584*, 393–397, doi:10.1038/s41586-020-2591-3.
- Fricke, H. A., B. K. Galton-Fenzi, C. C. Walker, B. I. D. Freer, L. Padman, and R. DeConto (2025), Antarctica in 2025: Drivers of deep uncertainty in projected ice loss, *Science*, *387*(6734), 601–609, doi:10.1126/science.adt9619.
- Frieler, K., P. U. Clark, F. He, C. Buizert, R. Reese, S. R. M. Ligtenberg, M. R. van den Broeke, R. Winkelmann, and A. Levermann (2015), Consistent evidence of increasing Antarctic accumulation with warming, *Nature Climate Change*, *5*(4), 348–352, doi:10.1038/nclimate2574.
- Geoffroy, O., D. Saint-Martin, D. J. L. Olivié, A. Voldoire, G. Bellon, and S. Tytéca (2013), Transient climate response in a two-layer energy-balance model. Part I: Analytical solution and parameter calibration using CMIP5 AOGCM experiments, *Journal of Climate*, *26*, 1841–1857, doi:10.1175/JCLI-D-12-00195.1.
- Getraer, B., and M. Morlighem (2025), Increasing the Glen-Nye power-law exponent accelerates ice-loss projections for the Amundsen Sea Embayment, West Antarctica, *Geophysical Research Letters*, *52*(7), e2024GL112516, doi:10.1029/2024GL112516.
- Glaude, Q., B. Noël, M. Olesen, M. R. van den Broeke, W. J. van de Berg, R. Mottram, N. Hansen, A. Delhasse, C. Amory, C. Kittel, H. Goelzer, and X. Fettweis (2024), A factor

- two difference in 21st-century Greenland Ice Sheet surface mass balance projections from three regional climate models under a strong warming scenario (SSP5-8.5), *Geophysical Research Letters*, *51*(22), e2024GL111,902, doi:10.1029/2024GL111902.
- Goldberg, D. N., M. Morlighem, and N. Gourmelen (2026), Recent observations of Thwaites Glacier, West Antarctica are consistent with high rates of loss in next 50 years, *Geophysical Research Letters*, *53*(5), e2025GL118,823, doi:10.1029/2025GL118823.
- Good, S. A., M. J. Martin, and N. A. Rayner (2013), EN4: Quality controlled ocean temperature and salinity profiles and monthly objective analyses with uncertainty estimates, *Journal of Geophysical Research: Oceans*, *118*, 6704–6716, doi:10.1002/2013JC009067.
- Green, K. C., and J. S. Armstrong (2015), Simple versus complex forecasting: The evidence, *Journal of Business Research*, *68*, 1678–1685, doi:10.1016/j.jbusres.2015.03.026.
- Hamlington, B. D., A. S. Gardner, E. R. Ivins, J. T. M. Lenaerts, J. T. Reager, D. S. Trossman, E. D. Zaron, S. Adhikari, A. Arendt, A. Aschwanden, et al. (2020), Understanding of contemporary regional sea-level change and the implications for the future, *Reviews of Geophysics*, *58*, e2019RG000,672, doi:10.1029/2019RG000672.
- Hamlington, B. D., D. P. Chambers, T. Frederikse, S. Dangendorf, S. Fournier, B. Buzanga, and R. S. Nerem (2022), Observation-based trajectory of future sea level for the coastal United States tracks near high-end model projections, *Communications Earth & Environment*, *3*, 230, doi:10.1038/s43247-022-00537-z.
- Hamlington, B. D., A. Bellas-Manley, J. K. Willis, S. Fournier, N. Vinogradova, R. S. Nerem, C. G. Piecuch, P. R. Thompson, and R. E. Kopp (2024), The rate of global sea level rise doubled during the past three decades, *Communications Earth & Environment*, *5*, 601, doi:10.1038/s43247-024-01761-5.
- Hirschfeld, D., D. Behar, R. J. Nicholls, N. Cahill, T. James, B. P. Horton, M. E. Portman, R. Bell, M. Campo, M. Esteban, B. Goble, M. Rahman, K. A. Addo, F. A. Chundeli, M. Aunger, O. Babitsky, A. Beal, R. Boyle, J. Fang, A. Gohar, S. Hanson, S. Karamesines, M. J. Kim, H. Lohmann, K. McInnes, N. Mimura, D. Ramsay, L. Wenger, and H. Yokoki (2023), Global survey shows planners use widely varying sea-level rise projections for coastal adaptation, *Communications Earth & Environment*, *4*, 102, doi:10.1038/s43247-023-00703-x.
- Jevrejeva, S., L. P. Jackson, A. Grinsted, D. Lincke, and B. Marzeion (2018), Flood damage costs under the sea level rise with warming of 1.5 °C and 2 °C, *Environmental Research Letters*, *13*(7), 074,014, doi:10.1088/1748-9326/aacc76.
- Jongman, B., P. J. Ward, and J. C. J. H. Aerts (2012), Global exposure to river and coastal flooding: Long term trends and changes, *Global Environmental Change*, *22*(4), 823–835, doi:10.1016/j.gloenvcha.2012.07.004.
- Joughin, I., B. E. Smith, and B. Medley (2014), Marine ice sheet collapse potentially under way for the Thwaites Glacier Basin, West Antarctica, *Science*, *344*, 735–738, doi:10.1126/science.1249055.

- Kopp, R. E., R. M. Horton, C. M. Little, J. X. Mitrovica, M. Oppenheimer, D. J. Rasmussen, B. H. Strauss, and C. Tebaldi (2014), Probabilistic 21st and 22nd century sea-level projections at a global network of tide-gauge sites, *Earth's Future*, *2*(8), 383–406, doi:10.1002/2014EF000239.
- Kulp, S. A., and B. H. Strauss (2019), New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding, *Nature Communications*, *10*, 4844, doi:10.1038/s41467-019-12808-z.
- Levitus, S., J. I. Antonov, T. P. Boyer, O. K. Baranova, H. E. Garcia, R. A. Locarnini, A. V. Mishonov, J. R. Reagan, D. Seidov, E. S. Yarosh, and M. M. Zweng (2012), World ocean heat content and thermosteric sea level change (0–2000 m), 1955–2010, *Geophysical Research Letters*, *39*, L10,603, doi:10.1029/2012GL051106.
- Li, Y., J. A. Church, T. J. McDougall, and P. M. Barker (2022), Sensitivity of observationally based estimates of ocean heat content and thermal expansion to vertical interpolation schemes, *Geophysical Research Letters*, *49*, e2022GL101,079, doi:10.1029/2022GL101079.
- Ligtenberg, S. R. M., W. J. van de Berg, M. R. van den Broeke, J. G. L. Rae, and E. van Meijgaard (2013), Future surface mass balance of the Antarctic ice sheet and its influence on sea level change, simulated by a regional atmospheric climate model, *Climate Dynamics*, *41*(3–4), 867–884, doi:10.1007/s00382-013-1749-1.
- MacAyeal, D. R., K. Mankoff, B. Minchew, J. Moore, and M. Wolovick (2024), Glacial climate intervention: A research vision, *Tech. rep.*, Climate Systems Engineering Initiative, University of Chicago, doi:10.15784/601797.
- Mankoff, K. D., X. Fettweis, P. L. Langen, M. Stendel, K. K. Kjeldsen, N. B. Karlsson, B. Noël, M. R. van den Broeke, A. Solgaard, W. Colgan, J. E. Box, S. B. Simonsen, M. D. King, A. P. Ahlstrøm, S. B. Andersen, and R. S. Fausto (2021), Greenland ice sheet mass balance from 1840 through next week, *Earth System Science Data*, *13*, 5001–5025, doi:10.5194/essd-13-5001-2021.
- Martin, D. F., S. B. Kachuck, M. Trevers, J. D. Millstein, S. L. Cornford, and B. M. Minchew (2026), Rates of sea-level rise are highly sensitive to ice viscosity parameters in model benchmarks, *AGU Advances*, *7*(2), e2025AV001,946, doi:10.1029/2025AV001946.
- Mengel, M., A. Levermann, K. Frieler, A. Robinson, B. Marzeion, and R. Winkelmann (2016), Future sea level rise constrained by observations and long-term commitment, *Proceedings of the National Academy of Sciences*, *113*(10), 2597–2602, doi:10.1073/pnas.1500515113.
- Meyer, C. R., K. L. P. Warburton, A. N. Sommers, and B. M. Minchew (2026), A model of water extraction from the subglacial hydrologic system under idealized conditions, *The Cryosphere*, *20*(5), 2659–2680, doi:10.5194/tc-20-2659-2026.
- Millstein, J. D., B. M. Minchew, and S. S. Pegler (2022), Ice viscosity is more sensitive to stress than commonly assumed, *Communications Earth & Environment*, *3*, 57, doi:10.1038/s43247-022-00385-x.

- Moorman, R., A. F. Thompson, and E. A. Wilson (2023), Coastal polynyas enable transitions between high and low West Antarctic ice shelf melt rates, *Geophysical Research Letters*, *50*, e2023GL104724, doi:10.1029/2023GL104724.
- Mouginot, J., E. Rignot, A. A. Bjørk, M. van den Broeke, R. Millan, M. Morlighem, B. Noël, B. Scheuchl, and M. Wood (2019), Forty-six years of Greenland Ice Sheet mass balance from 1972 to 2018, *Proceedings of the National Academy of Sciences*, *116*(19), 9239–9244, doi:10.1073/pnas.1904242116.
- Naughten, K. A., P. R. Holland, and J. De Rydt (2023), Unavoidable future increase in West Antarctic ice-shelf melting over the twenty-first century, *Nature Climate Change*, *13*, 1222–1228, doi:10.1038/s41558-023-01818-x.
- Noël, B., L. van Kampenhout, J. T. M. Lenaerts, W. J. van de Berg, and M. R. van den Broeke (2021), A 21st century warming threshold for sustained Greenland ice sheet mass loss, *Geophysical Research Letters*, *48*, e2020GL090471, doi:10.1029/2020GL090471.
- Otosaka, I. N., A. Shepherd, E. R. Ivins, N.-J. Schlegel, C. Amory, M. R. van den Broeke, M. Horwath, I. Joughin, M. D. King, G. Krinner, et al. (2023), Mass balance of the Greenland and Antarctic ice sheets from 1992 to 2020, *Earth System Science Data*, *15*, 1597–1616, doi:10.5194/essd-15-1597-2023.
- Poinar, K., J. E. Box, B. E. Smith, T. G. Askjaer, T. L. Mote, B. D. Loomis, B. C. Medley, K. D. Mankoff, and R. S. Fausto (2026), Greenland Ice Sheet, in *Arctic Report Card 2025*, NOAA, doi:10.25923/rbsh-t897.
- Purkey, S. G., and G. C. Johnson (2010), Warming of global abyssal and deep Southern Ocean waters between the 1990s and 2000s: Contributions to global heat and sea level rise budgets, *Journal of Climate*, *23*, 6336–6351, doi:10.1175/2010JCLI3682.1.
- Rahmstorf, S. (2007), A semi-empirical approach to projecting future sea-level rise, *Science*, *315*, 368–370, doi:10.1126/science.1135456.
- Rignot, E., Y. Xu, D. Menemenlis, J. Mouginot, B. Scheuchl, X. Li, M. Morlighem, H. Seroussi, M. van den Broeke, I. Fenty, et al. (2016), Modeling of ocean-induced ice melt rates of five west Greenland glaciers over the past two decades, *Geophysical Research Letters*, *43*, 6374–6382, doi:10.1002/2016GL068784.
- Rignot, E., B. Scheuchl, J. B. Barre, V. Brancato, L. Charrier, H. Chen, E. Ciraci, A. Dinh, S. Herreid, S. Jeong, X. Li, T. Mitchell, Y. Mohajerani, S. Shamsian, V. Tolpekin, I. Velicogna, and M. Wollersheim (2026), Thirty years of glacier grounding line retreat in Antarctica, *Proceedings of the National Academy of Sciences*, *123*(10), e2524380123, doi:10.1073/pnas.2524380123.
- Robel, A. A., G. H. Roe, and M. Haseloff (2018), Response of Marine-Terminating Glaciers to Forcing: Time Scales, Sensitivities, Instabilities, and Stochastic Dynamics, *Journal of Geophysical Research: Earth Surface*, *123*(9), 2205–2227, doi:10.1029/2018JF004709.

- Robel, A. A., H. Seroussi, and G. H. Roe (2019), Marine ice sheet instability amplifies and skews uncertainty in projections of future sea-level rise, *Proceedings of the National Academy of Sciences*, *116*(30), 14,887–14,892, doi:10.1073/pnas.1904822116.
- Rohde, R. A., and Z. Hausfather (2020), The Berkeley Earth land/ocean temperature record, *Earth System Science Data*, *12*, 3469–3479, doi:10.5194/essd-12-3469-2020.
- Rosier, S. H. R., G. H. Gudmundsson, A. Jenkins, and K. A. Naughten (2025), Calibrated sea level contribution from the Amundsen Sea sector, West Antarctica, under RCP8.5 and Paris 2C scenarios, *The Cryosphere*, *19*, 2527–2557, doi:10.5194/tc-19-2527-2025.
- Rounce, D. R., R. Hock, F. Maussion, R. Hugonnet, W. Kochtitzky, M. Huss, E. Berthier, D. Brinkerhoff, L. Compagno, L. Copland, D. Farinotti, B. Menounos, and R. W. McNabb (2023), Global glacier change in the 21st century: Every increase in temperature matters, *Science*, *379*, 78–83, doi:10.1126/science.abo1324.
- Schoof, C. (2007), Ice sheet grounding line dynamics: Steady states, stability, and hysteresis, *Journal of Geophysical Research: Earth Surface*, *112*, F03S28, doi:10.1029/2006JF000664.
- Schoof, C. (2010), Ice-sheet acceleration driven by melt supply variability, *Nature*, *468*(7325), 803–806.
- Seroussi, H., S. Nowicki, A. J. Payne, H. Goelzer, W. H. Lipscomb, A. Abe-Ouchi, C. Agosta, T. Albrecht, X. Asay-Davis, et al. (2020), ISMIP6 Antarctica: a multi-model ensemble of the Antarctic ice sheet evolution over the 21st century, *The Cryosphere*, *14*, 3033–3070, doi:10.5194/tc-14-3033-2020.
- Siegert, M., H. Sevestre, M. J. Bentley, J. Brigham-Grette, H. Burgess, S. Buzzard, M. Cavitte, S. L. Chown, F. Colleoni, R. M. DeConto, H. A. Fricker, E. Gasson, S. M. Grant, A. M. Gulisano, S. Hancock, K. R. Hendry, S. F. Henley, R. Hock, K. A. Hughes, D. Karentz, J. D. Kirkham, B. Kulesa, R. D. Larter, A. Mackintosh, V. Masson-Delmotte, F. S. McCormack, H. Millman, R. Mottram, T. A. Moon, T. Naish, C. Nath, B. Orlove, P. Pearson, J. Rogelj, J. Rumble, S. Seabrook, A. Silvano, M. Sommerkorn, L. A. Stearns, C. R. Stokes, J. Stroeve, and M. Truffer (2025), Safeguarding the polar regions from dangerous geoengineering: a critical assessment of proposed concepts and future prospects, *Frontiers in Science*, *Volume 3 - 2025*, doi:10.3389/fsci.2025.1527393.
- Slater, D. A., and F. Straneo (2022), Submarine melting of glaciers in Greenland amplified by atmospheric warming, *Nature Geoscience*, *15*(10), 794–799, doi:10.1038/s41561-022-01035-9.
- Slater, D. A., D. Felikson, F. Straneo, H. Goelzer, C. M. Little, M. Morlighem, X. Fettweis, and S. Nowicki (2020), Twenty-first century ocean forcing of the Greenland ice sheet for modelling of sea level contribution, *The Cryosphere*, *14*, 985–1008, doi:10.5194/tc-14-985-2020.
- Straneo, F., and P. Heimbach (2013), North Atlantic warming and the retreat of Greenland’s outlet glaciers, *Nature*, *504*(7478), 36–43, doi:10.1038/nature12854.

- Straneo, F., P. Heimbach, O. Sergienko, G. Hamilton, G. Catania, S. Griffies, R. Hallberg, A. Jenkins, I. Joughin, R. Motyka, et al. (2013), Challenges to understanding the dynamic response of Greenland’s marine terminating glaciers to oceanic and atmospheric forcing, *Bulletin of the American Meteorological Society*, *94*, 1131–1144, doi:10.1175/BAMS-D-12-00100.1.
- Tedstone, A. J., P. W. Nienow, A. J. Sole, D. W. F. Mair, T. R. Cowton, I. D. Bartholomew, and M. A. King (2013), Greenland ice sheet motion insensitive to exceptional meltwater forcing, *Proceedings of the National Academy of Sciences*, *110*(49), 19,719–19,724, doi:10.1073/pnas.1315843110.
- Tiggeloven, T., H. de Moel, H. C. Winsemius, D. Eilander, G. Erkens, E. Gebremedhin, A. Diaz Loaiza, A. Bouwman, J. van Huijstee, W. Ligtoet, S. Kuzma, T. Luo, C. Iceland, and P. J. Ward (2020), Global-scale benefit–cost analysis of coastal flood adaptation to different flood risk drivers using structural measures, *Natural Hazards and Earth System Sciences*, *20*, 1025–1044, doi:10.5194/nhess-20-1025-2020.
- Ultee, L., D. Felikson, B. Minchew, L. A. Stearns, and B. Riel (2022), Helheim Glacier ice velocity variability responds to runoff and terminus position change at different timescales, *Nature Communications*, *13*(1), 6022, doi:10.1038/s41467-022-33292-y.
- United Nations Environment Programme (2023), Adaptation gap report 2023: Underfunded. underprepared., *Tech. rep.*, UNEP, doi:10.59117/20.500.11822/43796.
- Wong, T. E., A. M. R. Bakker, K. Ruckert, P. Applegate, A. B. A. Slangen, and K. Keller (2017), BRICK v0.2, a simple, accessible, and transparent model framework for climate and regional sea-level projections, *Geoscientific Model Development*, *10*, 2741–2760, doi:10.5194/gmd-10-2741-2017.
- Wood, M., E. Rignot, I. Fenty, L. An, A. Bjork, M. Van Den Broeke, C. Cai, E. Kane, D. Menemenlis, R. Millan, M. Morlighem, J. Mouginot, B. Noël, B. Scheuchl, I. Velicogna, J. K. Willis, and H. Zhang (2021), Ocean forcing drives glacier retreat in Greenland, *Science Advances*, *7*(1), eaba7282, doi:10.1126/sciadv.aba7282.
- Xu, Y., E. Rignot, I. Fenty, D. Menemenlis, and M. Flexas (2013), Subaqueous melting of Store Glacier, west Greenland from three-dimensional, high-resolution numerical modeling and ocean observations, *Geophysical Research Letters*, *40*, 4648–4653, doi:10.1002/grl.50825.
- Zemp, M., et al. (2025), Community estimate of global glacier mass changes from 2000 to 2023, *Nature*, *639*, 382–388, doi:10.1038/s41586-024-08545-z, glaMBIE consortium.

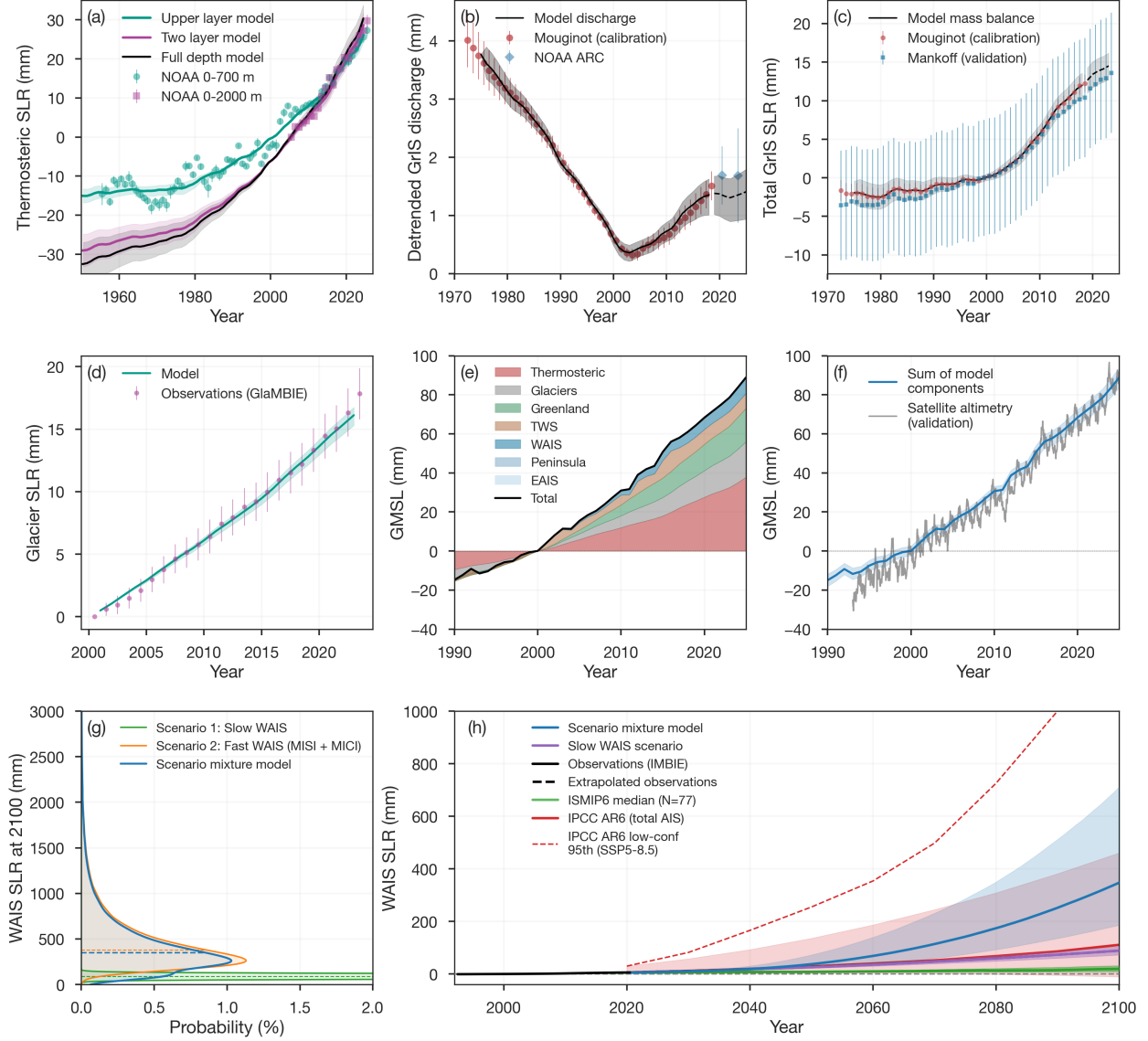


Figure 1: Model calibration, validation, and WAIS uncertainty. (a) Our two-layer energy balance model fitted jointly to NOAA thermosteric values at different depth layers with the full depth model used in the forecasts (Supplementary Materials). (b) Greenland ice discharge modeled as a cumulative time-delay response to subsurface ocean warming with the calibration data from (*Mouginot et al., 2019*) shown with filled circles and diamonds showing NOAA Arctic Report Card estimates (*Poinar et al., 2026*). (c) Greenland total mass balance with validation data shown in blue squares with 90% CI error bars (*Mankoff et al., 2021*). (d) Global glacier model fitted to GlaMBIE consensus observations (2000–2023) using a linear temperature-driven model. (e) Stacked per-component contributions to the temperature-forced hindcast, including EAIS. (f) Sum of components used for budget closure, including EAIS (see text), compared with satellite altimetry (cf. Fig. 7); shaded band is the 5–95% Monte Carlo range. (g) WAIS scenario distributions at year 2100; dashed lines indicate medians. (h) WAIS projections from our full scenario mixture (blue) and slow WAIS scenario (purple) compared with observations (black, *Otosaka et al., 2023*), the ISMIP6 ensemble (green; *Seroussi et al., 2020*), and IPCC AR6 projections (red).

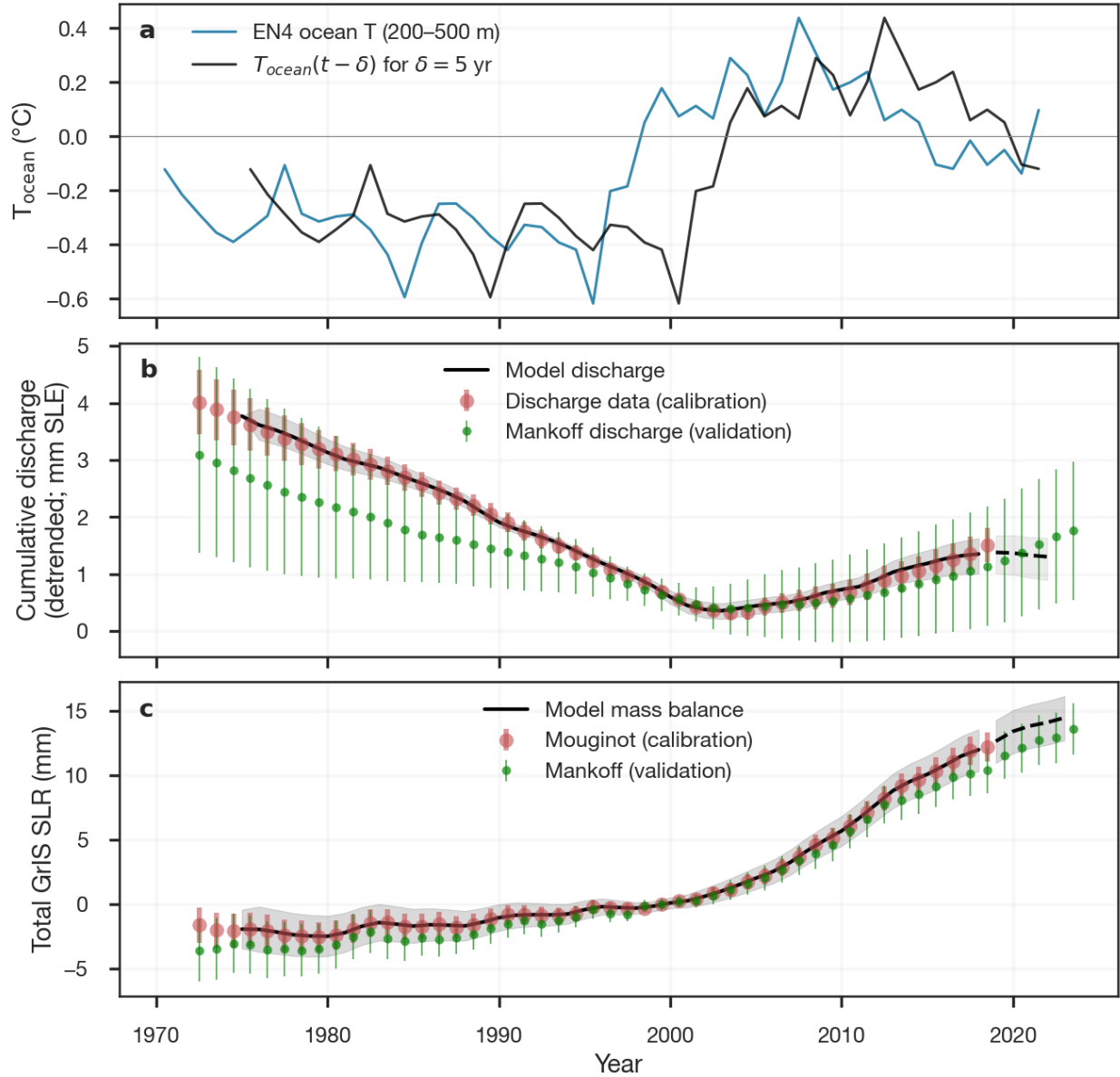


Figure 2: **Greenland discharge delay model fit.** (a) EN4 subsurface ocean temperature (200–500 m, annual) and the delayed signal $T_{\text{ocean}}(t - \delta)$ with $\delta = 5$ yr. (b) Detrended cumulative Greenland ice discharge showing model fit (black) with 90% posterior CI, compared with the calibration (*Mouginot et al.*, 2019) and validation data (*Mankoff et al.*, 2021) in red and green, respectively. (c) Total Greenland mass balance including both SMB and discharge.

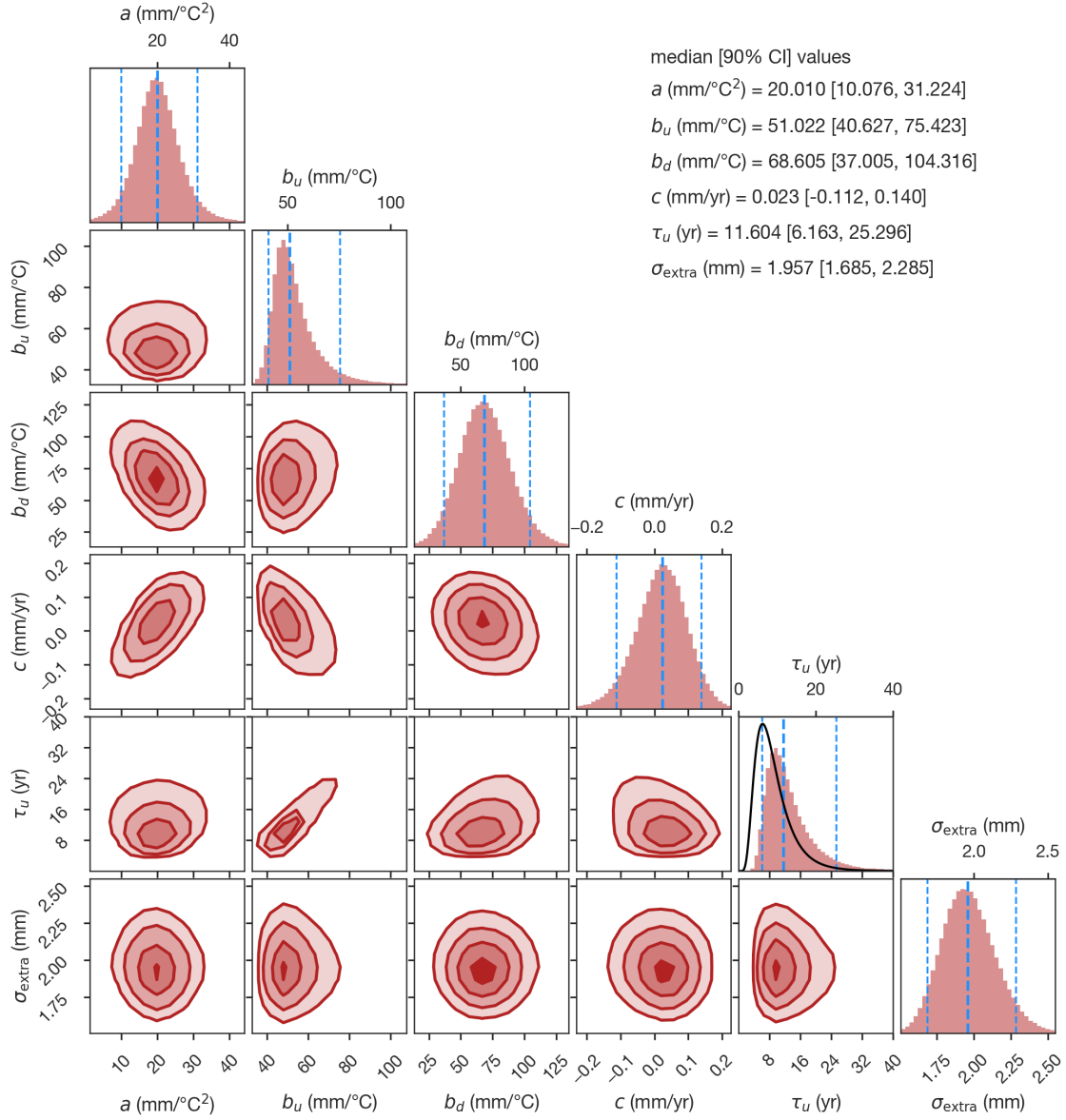


Figure 3: **Two-layer thermosteric model posterior correlation structure.** The joint posterior for the six parameters of the two-layer energy balance model fitted to NOAA 0–700 m and 0–2000 m thermosteric sea level data. Diagonal panels show marginal posterior histograms (red), with dashed vertical lines marking the median and 90% credible interval. Off-diagonal panels show the joint posterior density, with contours denoting 1, 2, and 3 standard deviations. Only τ_u has an informative prior (black curve) while the other five parameters have flat, uninformative priors (not shown).

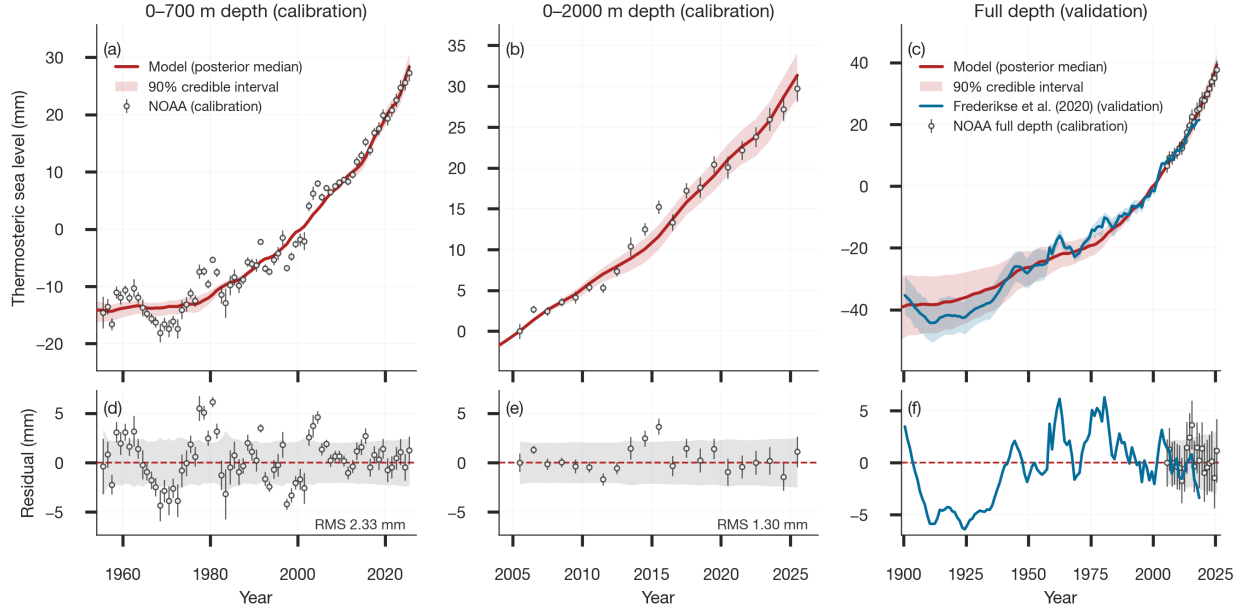


Figure 4: **Two-layer thermosteric model: fit and validation.** (a) Posterior median of the upper layer model H_u (Eq. 8) and 90% credible interval fitted to NOAA 0–700 m observations (circles with error bars). (b) As in (a), for the 0–2000 m record and H_{ud} (Eq. 11). (c) Validation of the full-depth model H_{full} (Eq. 12) against the independent *Frederikse et al. (2020)* reconstruction (blue), withheld from calibration. Also shown for reference are the NOAA-derived full-depth values (circles). (d–f) Residuals corresponding to (a)–(c), with the gray band showing the 90% credible interval on the residual and the RMS residual reported in each calibration panel. RMS values relative to *Frederikse et al. (2020)* reconstruction are 2.5 mm for the observational period 1957–2018 and 3.8 mm pre-1957 when the reconstruction is constrained by models rather than direct observations.

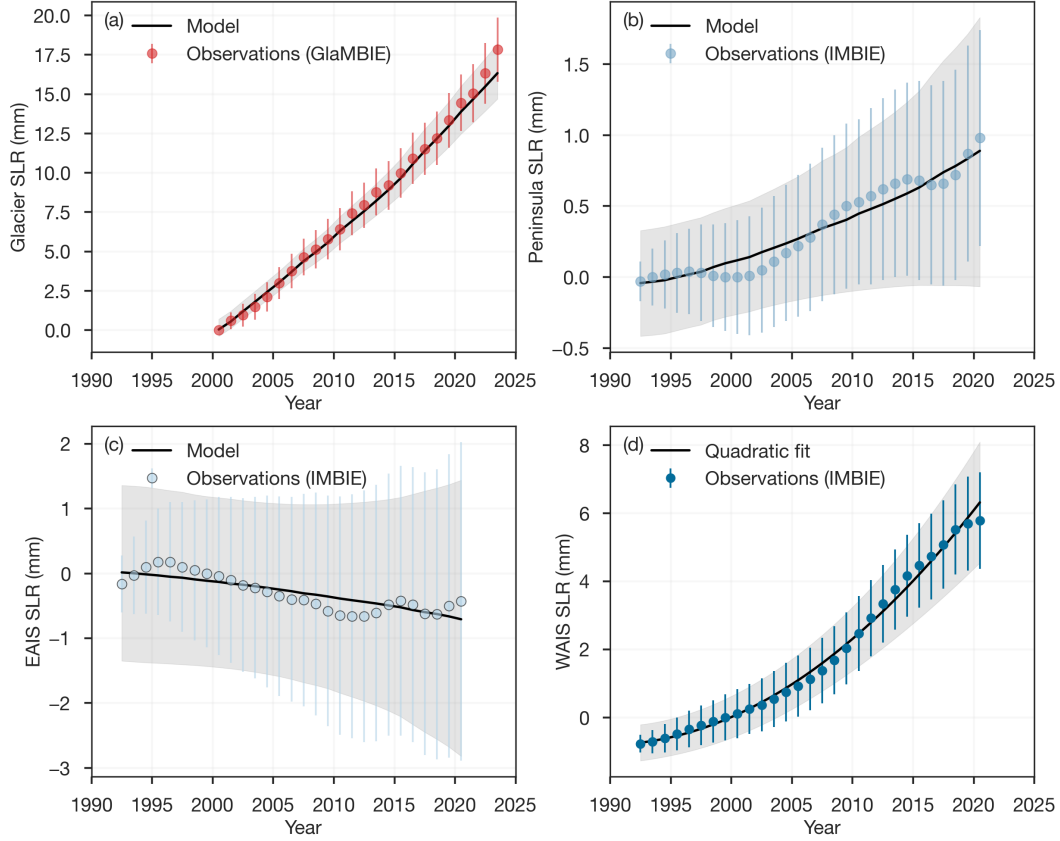


Figure 5: **Per-component calibration fits.** (a) Glaciers, (b) Antarctic Peninsula, and (c) EAIS: posterior median rate-temperature model (black) and 90% credible interval (gray) against GlaMBIE/IMBIE observations (circles, $\pm 2\sigma$). (d) WAIS: descriptive quadratic-in-time fit (black).

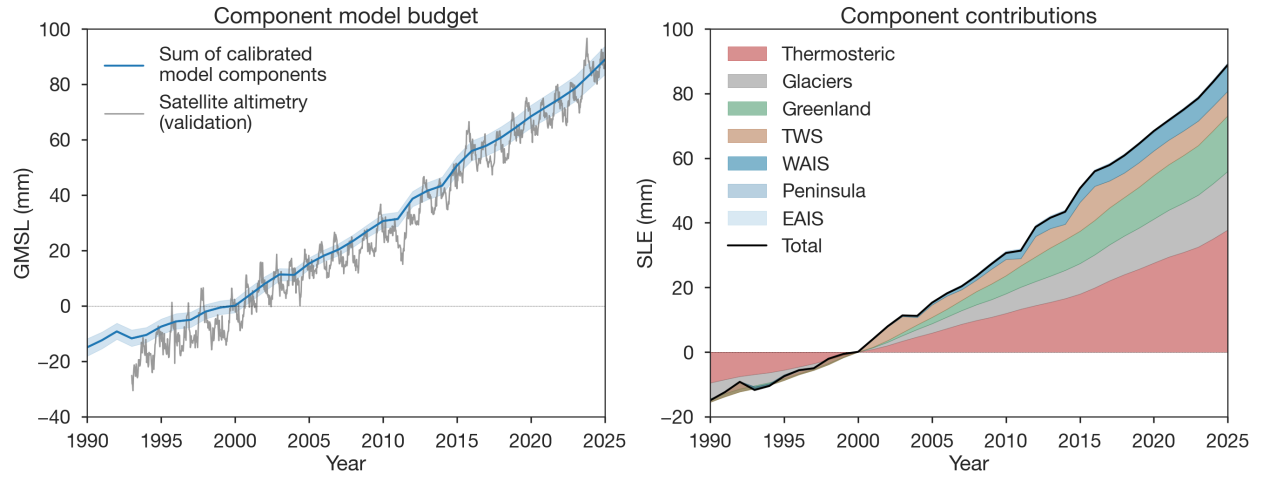


Figure 6: **Budget closure: sum of model components versus observed GMSL.** (a) Sum of components (blue) and NASA satellite altimetry (grey) with shaded band showing the 90% confidence intervals (*Beckley et al.*, 2017; *Hamlington et al.*, 2020). (b) Stacked per-component contributions and their sum in black.

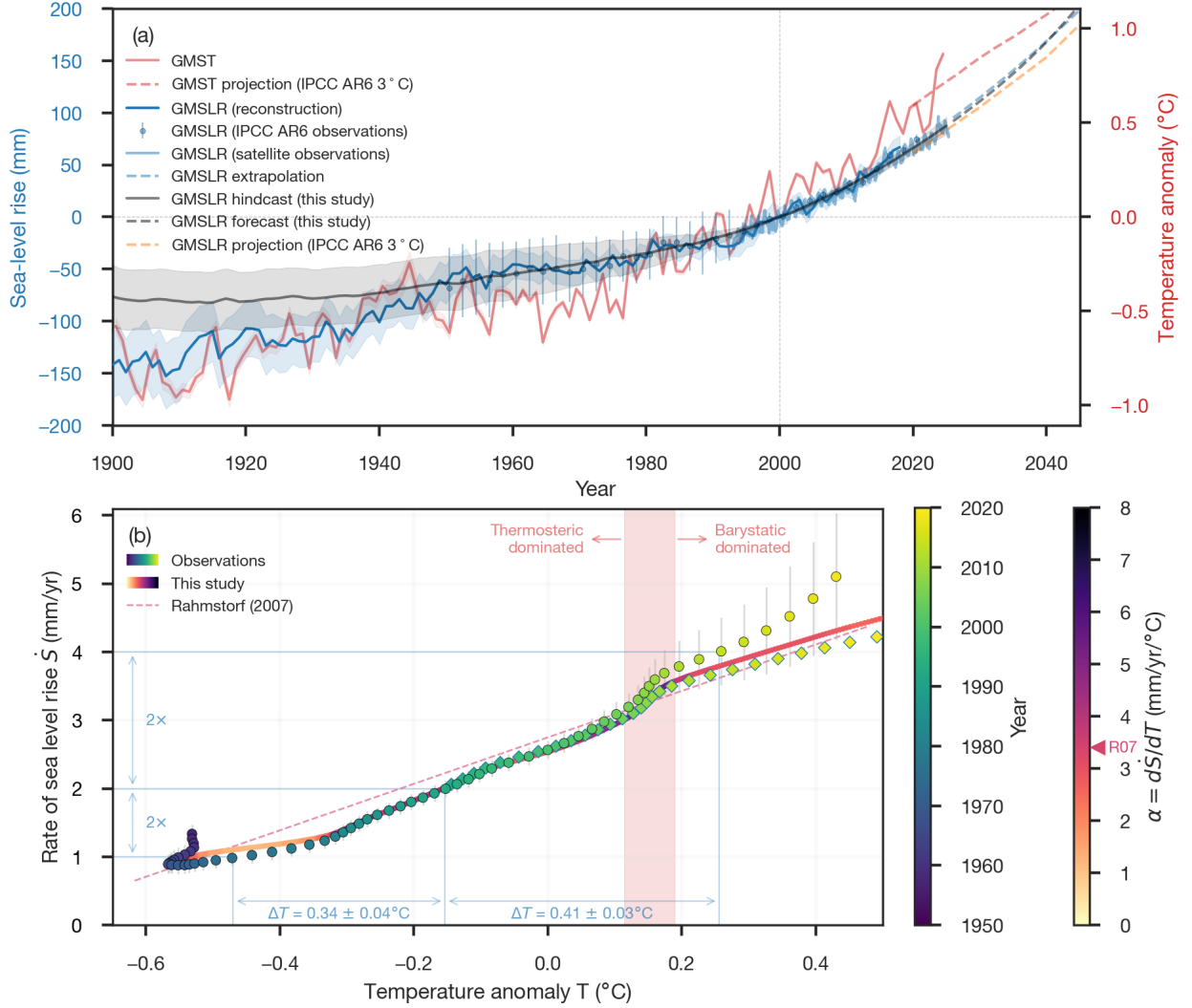


Figure 7: **Observed global mean sea-level rise and global mean surface temperature.** (a) GMSL reconstructions (blue line *Frederikse et al.* (2020); blue circles *Fox-Kemper et al.* (2021)) along with more recent NASA satellite altimetry (light blue; *Beckley et al.* (2017)) show sea-levels rising with warming. Hindcast results of this study (grey line with 90% CI shading) show good agreement back to 1950, the earliest year when GMSL reconstructions agree (*Fox-Kemper et al.*, 2021). Different projection methods (dashed lines) show the divergence of IPCC AR6 projections and observed trends within a decade. GMST (red; (*Rohde and Hausfather*, 2020)) is referenced to the right hand axis. (b) Rate of GMSL rise ($\dot{S} = dGMSL/dt$) versus GMST anomaly, T . Observed values are shown in circles (*Frederikse et al.*, 2020) and diamonds (*Beckley et al.*, 2017), colored by year, and the component model of this study is a line colored by instantaneous transient sensitivity $\alpha = d\dot{S}/dT$. The Rahmstorf (2007) semi-empirical model is shown in a color corresponding to its α value (*Rahmstorf*, 2007). The vertical red band indicates the shift in the primary contributor from ocean thermal expansion to changes in glacier ice and water storage (barystatic). The sinuous tail in the early observations is largely due to dam impoundment of water rather than a thermodynamic signal (*Frederikse et al.*, 2020).

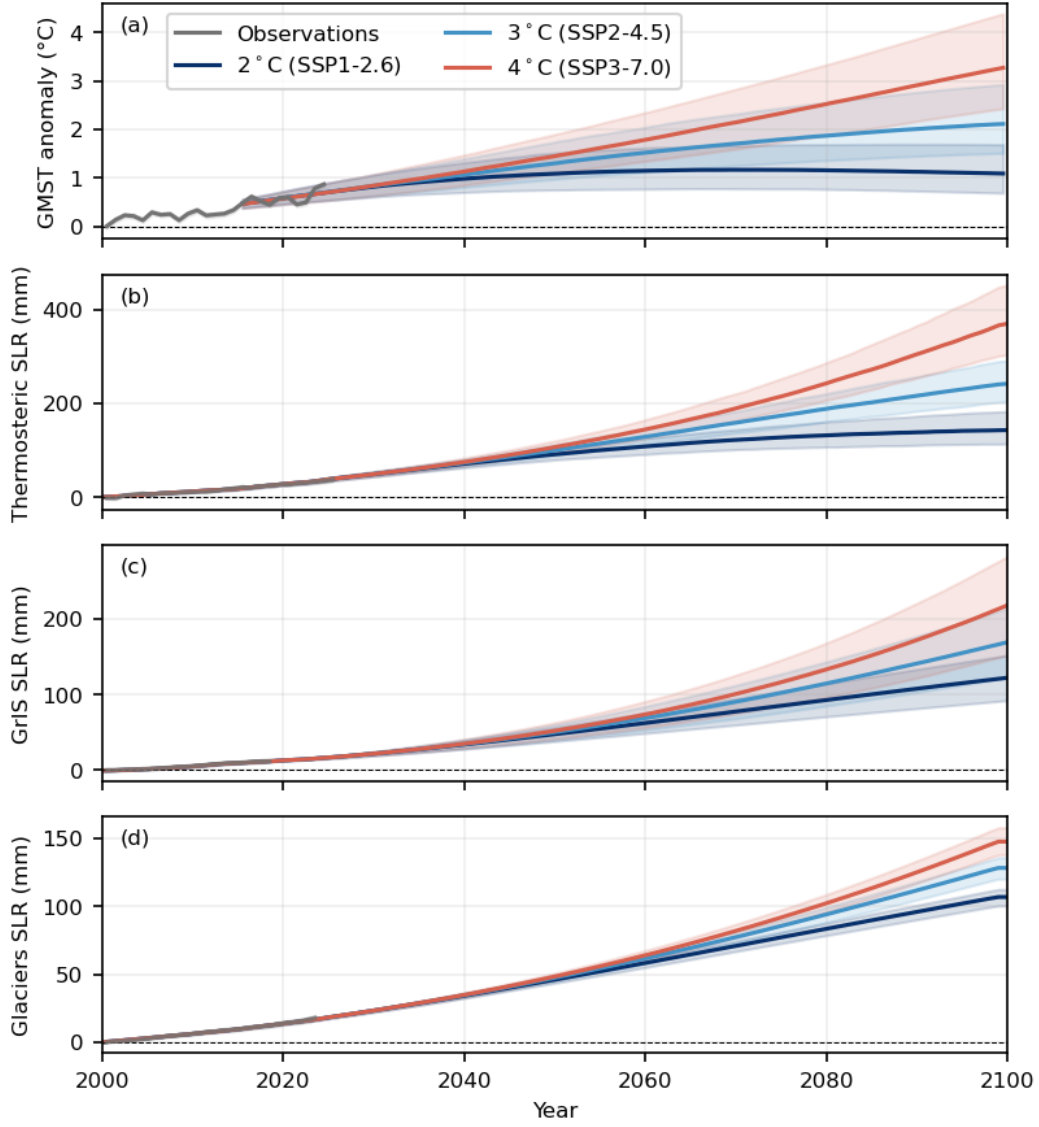


Figure 8: **Component-level forecasts under three emissions scenarios.** (a) Observed global mean surface temperature (GMST) anomaly (grey) relative to 2000, with SSP1-2.6, SSP2-4.5, and SSP3-7.0 projections (IPCC AR6) and their 90% CI (5–95th percentile, shaded). The SSP projections are labeled in the legend by their median 2100 anomaly relative to pre-industrial temperatures to provide both pre-industrial and 21st century baselines. (b) Thermosteric sea-level contribution: NOAA observations (grey) and forecast median with 90% CI from the full depth model driven by the corresponding GMST trajectory. (c) Greenland Ice Sheet (GrIS) contribution: observed cumulative mass balance (grey) and scenario forecasts from the discharge and SMB models. (d) Global glacier contribution: observations (grey) and scenario forecasts from the temperature-driven glacier model. Antarctic Peninsula and EAIS are excluded from this plot because their contributions are within the uncertainties of the components shown.

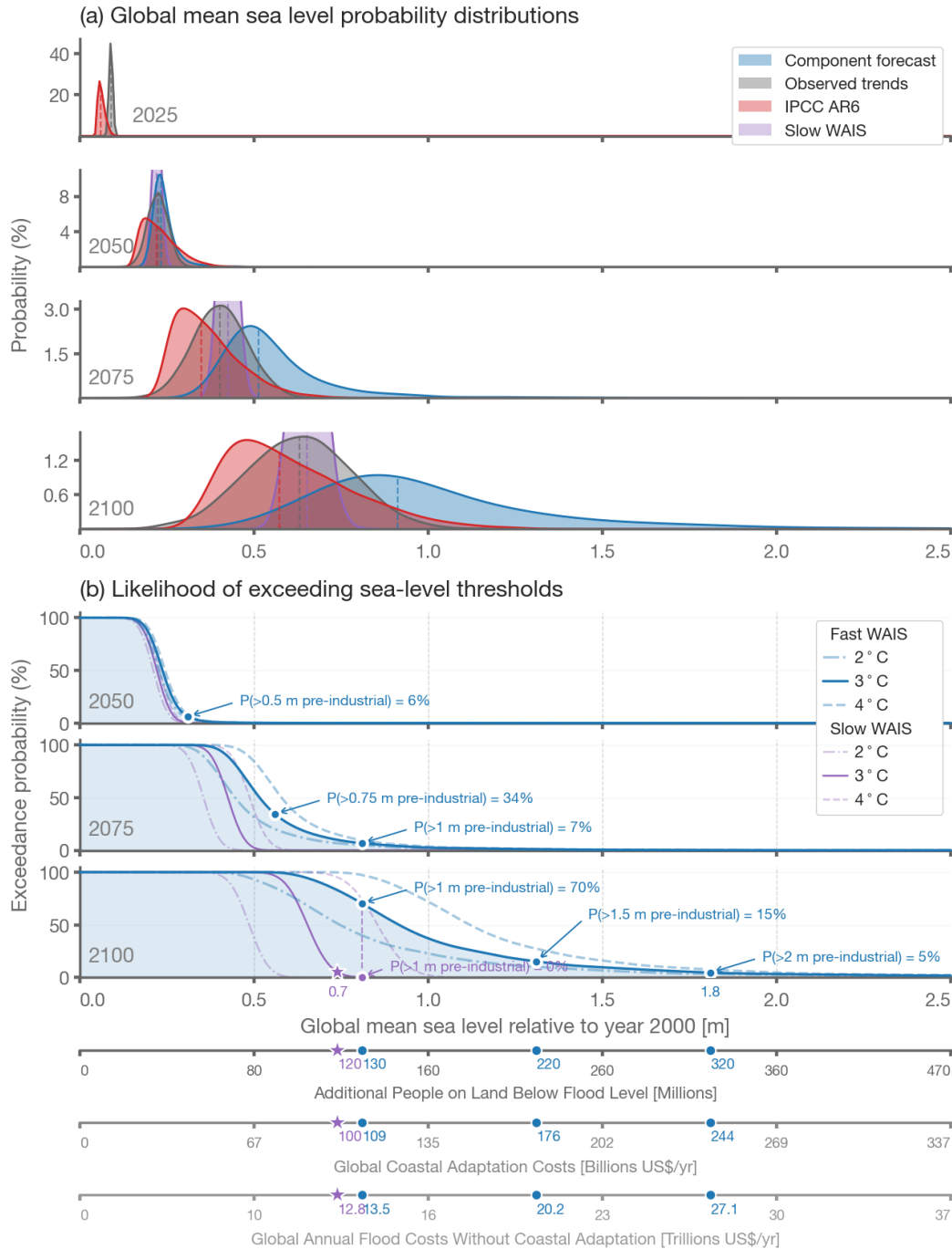


Figure 9: Global mean sea-level probabilities. (a) Probability distributions for different forecasting approaches and the IPCC AR6 projections. The ‘slow WAIS’ scenario (purple, clipped for figure clarity) assumes mass loss does not accelerate outside of the observed bounds (Scenario 1) while the component forecast (blue curve) allows for accelerated WAIS mass loss. Vertical dashed lines indicate the respective medians. (b) Probability of exceeding a given sea-level threshold for different warming trajectories including the WAIS scenario mixture model that allows for fast mass loss (blue) and the slow WAIS scenario (purple). Lower axes show estimates of the corresponding number of people currently living on land that will be below flood level (*Kulp and Strauss, 2019*) and financial costs (*Tiggeloven et al., 2020; Jevrejeva et al., 2018*) (Supplementary Material). Purple star indicates the 5% exceedance probability for the case that WAIS.

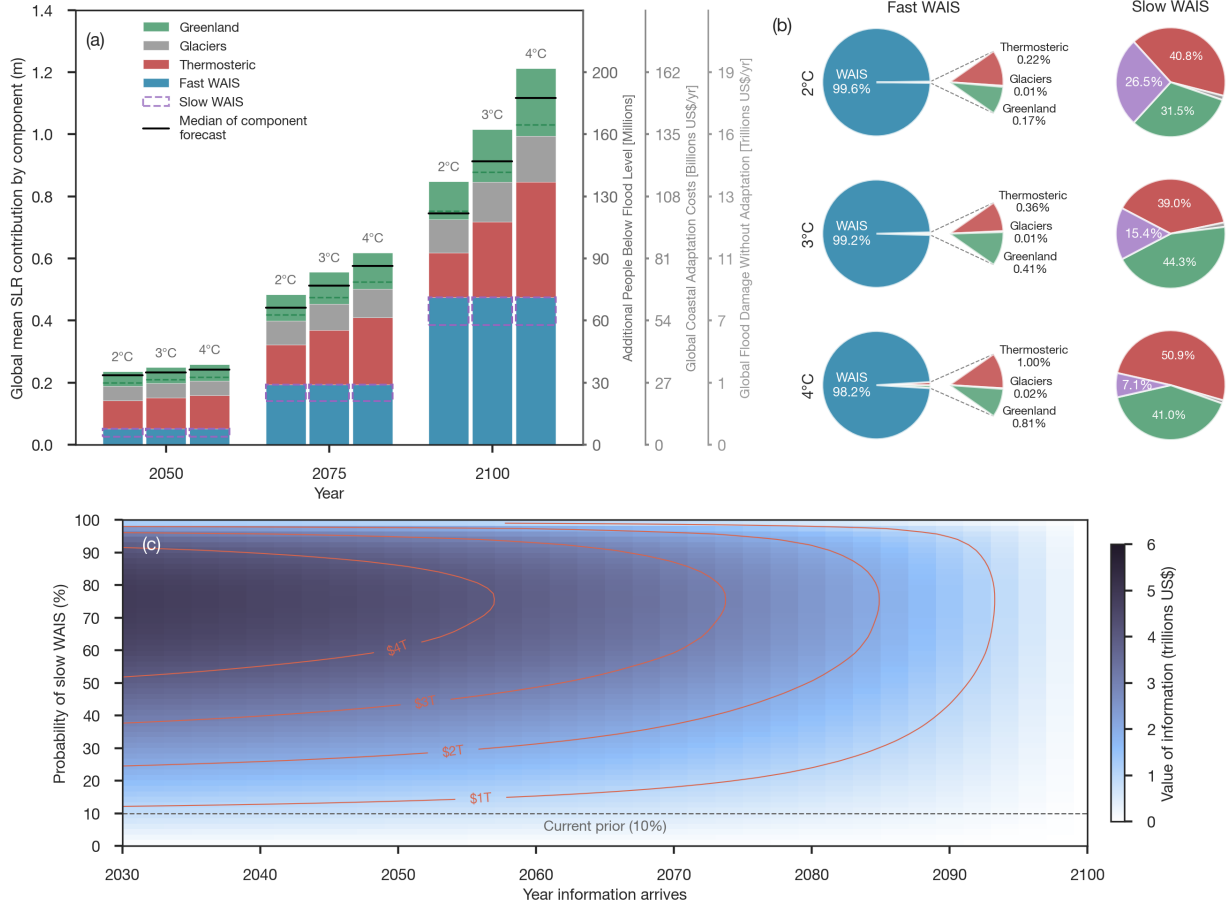


Figure 10: **Component contributions, variance decomposition, and value of information.** (a) 21st century mean GMSLR contribution by component and respective societal costs. Green dashed lines within Greenland bars partition discharge anomaly (below) from SMB anomaly (above). The purple box shows how much WAIS will contribute if it loses mass slowly while the rest of the blue bar shows the potential SLR from rapid ice loss. (b) Variance-fractions at 2100. WAIS accounts for approximately 99% of total forecast variance when rapid ice loss is possible. Fan breakouts show the non-WAIS contributions. If WAIS loses mass slowly, the variance distributions fundamentally shift (right column) (c) Financial value of resolving WAIS uncertainty as a function of when information arrives. Contours and colormap in trillions US\$ based on coastal adaptation costs (Supplemental Material). The plot shows that information has more value if it arrives early enough to inform key decisions and if it provides sufficient confidence that WAIS will lose mass slowly, allowing planning around lower future sea levels.